Review

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Recent Development of Morphology-Controlled Hybrid Nanomaterials for Triboelectric Nanogenerator: A Review

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Abstract: Being cognizant of modern electronic devices, the scientists are continuing to investigate renewable green-energy resources for a decade. Amid different energy harvesting systems, the triboelectric nanogenerators (TENGs) have been found to be the most promising mechanical harvesting technology and have drawn attention to generate electrical energy. Thanks to its instant output power, choice to opt for wide-ranging materials, low maintenance cost, easy fabrication process and environmentally friendly nature. Due to numerous working modes of TENGs, it is dedicated to desired application at ambient conditions. In this review, an advance correlation of TENGs have been explained based on the variety of nanostructures, including 0D, 1D, 2D, 3D, metal organic frameworks (MOFs), coordination polymers (CPs), covalent organic frameworks (COFs), and perovskite materials. Moreover, an overview of previous and current perspectives of various nanomaterials, synthesis, fabrication and their applications in potential fields have been discussed in detail.

Keywords: Nanogenerators, Triboelectric, Harvesting, Mechanical Energy, Flexible Devices

1. Introduction

1.1. History and Background

Energy harvesters relates the proceeding of sustainable energy resources which has become more desirable and minimizing energy transmission loss. It is distinct from traditional resources which does not require any specific fuel such as coal, gas or oil. The available surrounding energies with no polluting gas can be a future option for generating the power to the human society.[1a,b] In order to generate electrical power, the energy harvesters utilize numerous kinds of resources from the environment including thermal, radiant, and mechanical energy.[1c] The mechanical energy is ever present in our daily routine and can be utilized as source for triboelectrification which results from the contact of two different materials.[2a,b] Based on industrial point of view, it is commonly acknowledged as negative effect because the electrostatic charges prompted ignition, explosions dust, dielectric disintegration, damage of electronic components etc. From the perspective of energy, the electrostatic charges establish a capacitive energy device once the two different triboelectric surfaces are isolated, it has been concluded that the creation of early electrostatic generators in particular friction machine and Van de Graaf generator.[2c,d] Numerous innovative approaches have come into existence to extract energy from waste or moving towards more sustainable energy source. The term 'TENG' its detailed form is triboelectric nanogenerator which convert ambient wasted mechanical energy into valuable electrical energy and have drawn attention of scientific community since its invention.[2a,3a] The TENG was developed by Wang et al. in 2012 and the conversion is entirely depended on the tribo-

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electric effect and electrostatic induction.[1c,2d,3b,c] Subsequently, the TENGs dominates the benefits that comprise higher energy conversion efficiency, output efficiency, facile fabrication, minimal cost, lightly built, broad range of materials, and different working modes.[3c–e] As per the available reports, various publications have been demonstrated the effectiveness of TENG and as well as the essentiality for the coming future and its capabilities by taking into account. More particularly, the contact/triboelectric electrification offers static polarized charges on the surface of materials which is in contact, whereas electrostatic induction pushes the mechanical energy transformation into electric power via electrical potential deviation inspired by mechanical agitated separation.[4a–c] During a last decade, the TENG idea has been augmented to various working modes to empower wide spread application framework, that includes mechanical trembling, human activity, wind, and water waves.[5a,b] Presently, the TENG can be utilized in numerous applications such as self-powered buoy for navigation safety, wearable sensor, trajectory-based tracking, wireless energy consumption, seashore IoT (internet of things) as blue energy harvester, health monitoring, intelligent sports etc.[5c–f] The maximum power density and power conversion efficiency has been achieved tens of mW/cm^2 level and 85 %, respectively.[6a,b] The electrification effect persists two diverse materials, primarily one is polymer and another one is organic materials to develop flexible and stretchable TENG devices. To emphasize the friction area and ensure maximum output efficiency, the micro/nano patterned array has been accepted.[7a,b]

Numerous TENG based reviews are reported based on the working principal modes, phenomenon, and its applications. Wang et al. assessed the TENG application in sparking of an energy and sensor evolution.[1b] Luo et al. assessed the TENG based innovative technology towards intelligent sports.[5f] Yi et al. assessed the advancement of TENG based health monitoring.[5e] Kim et al. assessed the structure, mechanism and applications of TENG based materials.[1c] Wu et al. assessed, the TENG is foundation of energy of new era and discussed applications in detail.[2d] Zhang et al. assessed and anchored the advancement of inorganic materials for TENG.[3b] Zhang et al. elaborated theoretical comparison, equivalent transformation, and conjunction operations of electromagnetic induction (EMI) generator and TENG for harvesting and mechanical energy.[8] Pan et al. assessed triboelectric and piezoelectric TENG for future soft robots and machines.[9]

In this review, we tried to demonstrate the basic concept of TENG and elaborated numerous dimensional materials varying from 0D to 3D that concludes the robust structure such as MOFs (Metal Organic Frameworks), CPs (Coordination Polymers), and COFs (Covalent Organic Frameworks), Perovskite materials for the improvement of performance. In addition, apart from its usage in perovskite and 0D to 3D nanomaterials is very rare and primarily focused to bring these literatures in one review. This review is thoroughly debated on various nanomaterials with their synthesis method and applications for TENG (Scheme 1).

1.2. Working Principle of TENG

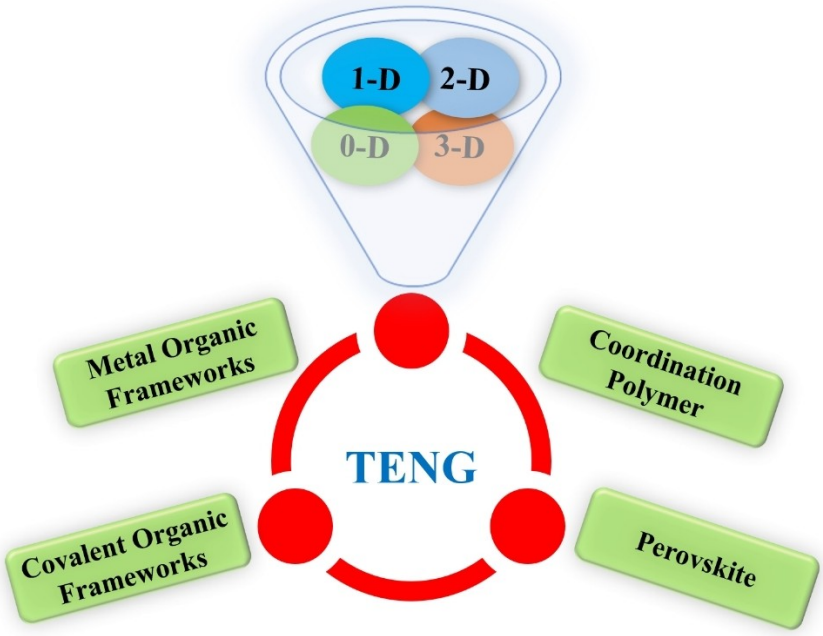
The working principle of TENG is entirely based on the contact electrification and electrostatic induction which is coupled together. To understand the fundamental procedures of TENG, four different modes have been discussed that includes (i) Vertical Contact-Separation (VCS) Mode (ii) Lateral Sliding Mode (iii) Single Electrode (SE) Mode, and (iv) Freestanding Triboelectric-Layer Mode (FSTL).[10,11]

1.2.1. Vertical Contact-Separation Mode

The TENGs working phenomenon based on the periodic change of potential deviation that occurs due to repeated separation and re-connect of opposite charges within the inner surface of two different sheets. When the mechanical force is employed on the TENG device, the two sheets get close to each other as well as their internal structure. Due to this coupling, the charge transportation will occur with one side as +Ve charge and other one as -Ve charges. This process is called triboelectric effect. Further, when the deformation is relived, the two different surfaces (with opposite charges) will get separated inherently, therefore these left opposite charges will produce electric charge carrier (by introducing potential deviation) from top to bottom electrodes. To scrutinize the potential deviation, the charge carrier will be flowing from one electrode to other via external electrode. However, the electricity production via this approach will be access until the potential of the two electrodes will get back to zero again. Whenever, two sheets compressed against each other, the triboelectric will caused potential deviation that will begin to reduce to zero, therefore transported charge will flow back via external load by creating another current in opposite side. The alternating current (AC) pulse will be simultaneously produced when the periodic mechanical force lasts longer.[11–13a] As the two different materials are getting in contact for producing charges, one of them should be non-conducting, therefore the triboelectric charge will not be conducted but will remain within the inner surface of the sheet. Subsequently, these immobile triboelectric charges can produce AC flow in the external load under the influence of periodic deformation. The phenomenon is well explained in Figure 1**(a)**.

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Scheme 1. Schematic representation of different materials for TENG application

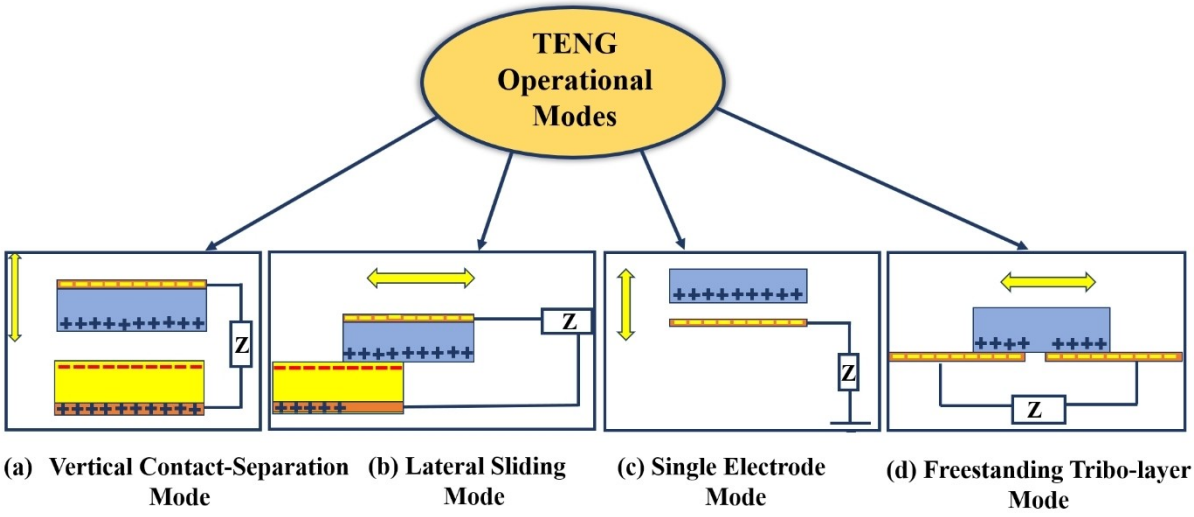


Figure 1. Operational modes of TENGs (a) Vertical contact-separation mode, (b) Lateral sliding mode, (c) Single electrode mode, and (d) Free-standing triboelectric.[13b] [Inspired from ref. 13b]

1.2.2. Lateral Sliding Mode

In this mode, the two main friction processes exist such as normal contact and lateral sliding. In Figure 1**(b)**, the TENG is designed and store the charge via in-plane sliding within two sheet surfaces in lateral direction. The tribo-electrification occurred due to sliding friction which causes periodic deformation within the two surfaces that creates potential deviation and flow of charge carriers in the external load. In equilibrium position, the surface of two polymeric materials is in contact or overlap to each other intimately. The materials ability to generate electrons due to large periodic deformation, where the tribo-electrification will permit one surface accumulated with +Ve charge and other one accumulated to -Ve

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charges by maintaining overall charge density. Within two tribo- surfaces, one will act as insulator that will distribute the charge in the surface layer and there will no leakage for prolonged time. Therefore, the charge separation within the + Ve and -Ve charged surface is minimum during overlapped stage, hence there will be little potential drop against the two electrodes. When the top sheet (with positively charged) begins to slide outward, the in-plane charge separation is start generating because of the reduced contact surface area. This separation will generate electric field from right to left parallel to plates by creating higher potential at top electrode.[11,14,15]

1.2.3. Single Electron Mode

This mode is more feasible and utilized to design fingertip-driven TENGs. The mechanism is demonstrated in Figure 1**(c)**, where conjunction of contact electrification and electrostatic phenomenon is shown. The interconnection of electrodes in sliding and separation contact mode has designed which may suffer some limitations. As an example, the mechanically movement energies have drawn due to walking on a floor, automobiles movement, finger tips or finger slumping on the touch screen devices where one friction layer is effortlessly moving so that electrodes cannot be interlinked. To surpass this issue, the SE mode TENG has been inducted for device fabrication. The single friction layer with rear electrode connected to the ground has been demonstrated in figure. However, the exterior object (either contacts or slide) on the TENG depicts as complementary friction layer. Due to periodic mechanical deformation energy as a consequences of contact separation/relative sliding concludes an AC in between the electrode and ground.[11,16,17]

1.2.4. Free-Standing Triboelectric Layer

This mode is quite similar to SE mode, where TENGs based device harvest mechanical energies via disturbance in objects. The mechanism of FSTL mode is illustrated in Figure 1**(d)**. In fact, the FSTL having freely moving objects.[18] Subsequently, the TENG configuration comprise pairs of interconnected electrodes whose dimensions should be such that free-standing layer that can cover whole the system.[6b,19] This free-standing layer gets in contact with electrodes, the current carrier flow in between them due to electrostatic field. The sliding on the electrodes can be planar and rotational manner.[18,19] The configuration of laterally aligned electrodes can be replaced with free-standing layer (vertical configuration) which is movable in between them. However, free-standing layer is conductive in nature then TENG needs dielectric layer in the underneath of electrode.[20] The FSTL can also function in the non-contact mode. In fact, if the triboelectric layers are energized with charge carrier by initial contact, then after contact sliding is not mandatory any longer. Even after, adequate separation process, the free-standing layer can be slide over the electrodes. To extend the lifetime of non-contact TENG, by decreasing the surface wearing because of sliding motion.[18]

2. Nanostructures Based TENGs Performance

The TENG based devices is primarily dependent on the nanostructures of different triboelectric nanomaterials. This review has thoroughly discussed various nanomaterials with their synthesis method and applications. We will mainly discuss about the different dimensions-based nanomaterials (0D, 1D, 2D, and 3D), metal-organic frameworks (MOFs), covalent organic frameworks (COFs), co-ordination polymers (CPs) and perovskite materials. Overviews of different nanomaterials are explained step by step with their TENG applications.

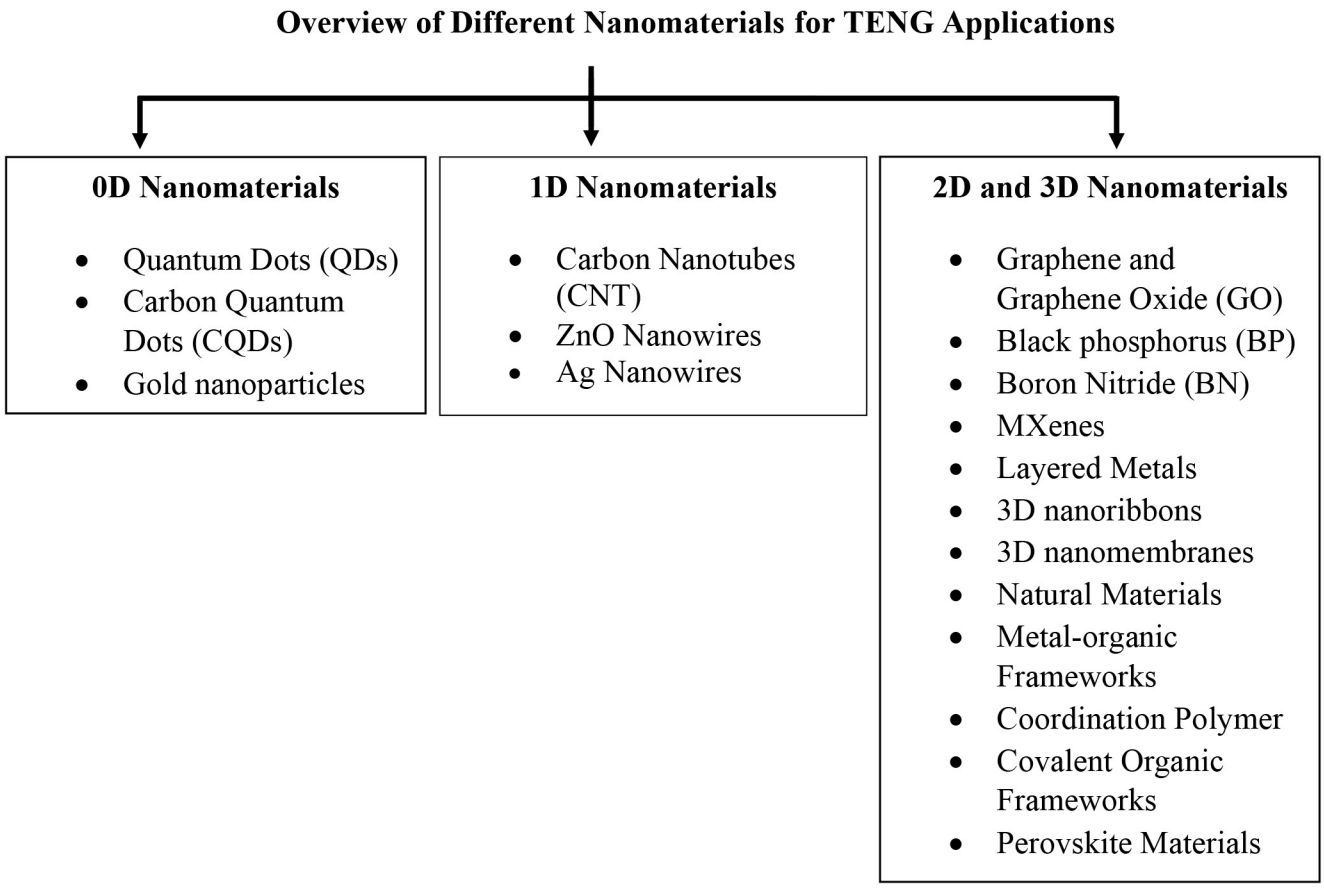
2.1. 0D Nanomaterials

In today's technological era, flexible electronics are gaining popularity for prominent applications such as electronic communication gadgets, biomedical health monitoring devices and display technology. Such research has encouraged in development of the flexible electronic devices with the use of zero-dimensional nanomaterials.[21–23] The zero-dimensional (0D) nanomaterials exhibit unique blend of properties which are primary function of their shape and size and some of them include quantum confinement effect observed in quantum dots[24,25] and surface plasmon resonance[24,26] in noble metals nanoparticles.[24] In order to fabricate the flexible electronic devices based on 0D nanomaterials, several fabrication methods and patterning methods were developed.

To achieve the desired shape and thickness, the bottom-up approach has found to be more suited. The methods comprise namely casting,[27] dip coating,[28] Langmuir-Blodgett deposition (LBD)[29] and, hence subsequently explored to develop 0D system. Additionally, different pattering methods have been explored that includes screen,[30,31] inkjet[32,33] and 3D printing[34,35] to design functional devices. The fabricated thin-film may offer subsequent applications in various fields like light emitting diode (LEDs) and solar cells stage. On the other hand, the nanoparticles (NPs) have been utilized as mechanical sensing elements and electrodes. Although, solar cells utilized NPs on its panel to capture the light/transportation of electrons. For the proper functionalities, the NPs are sandwiched between the different types of organic and inorganic layers. The final aim is to produce high quality device which concludes the desired shape, and their layout,

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resolution to maximize the efficiency advantages.[24] Numerous kind of 0D nanomaterials are discussed in details.

2.1.1. Quantum Dots (QDs)

QDs properties are interestingly different (optical and electronics) from the micron sized particles and bulk materials. It is having semiconductor characteristics with quite small zero-dimensional size about 2–10 nm. Based on the material properties of the quantum dots, including geometrical parameters allows QDs to emit electromagnetic waves with specific frequencies when irradiated by light or electromagnetic rays. Colloidal quantum dots are popular among researchers because of their high photoluminescence and ability to control their absorption and emission properties because of size dependent optoelectronic properties. Semiconductors from II–IV, III–V and IV–VI groups are usually used for making colloidal quantum dots where Si and Ge belongs to group IV; GaN, GaP, GaAs, InP corresponds to III–V groups and ZnO, CdS, CdSe belongs to II–VI group. Photoluminescence was shown by the semiconductor particles of such small size when single wavelength is emitted by stimulating UV light and it is possible to explain this photoluminescence. This can be explained via the quantum confinement effect. Quantum confinement effect explains the dynamics of semi-conductor particle whose diameter is smaller than the Bohr radius because of which excitons (quasiparticle) are squeezed in a complete 3D trap.[24,36] The quasiparticle is a system which is considered similar to that of hydrogen like system comprising of an electron and a hole where Bohr model predicted the distance between electron and hole around 1–10 nm for most of the semiconductors.

The electronics structure of these quantum dots turns out to be intermediate path between the localized and delocalized bands. The best suited example for QDs is like bound particle (particle in a box), because the excitor diameter is larger than spatial dimensions of QDs. The conclusion is that, the bandgap energy (BGE) is primarily depending upon the size of the QDs. When the absorption shifts to higher energy i.e blue shifts, it indicates that the particle size is decreasing and indicating the increase in the BGE. The adequate band gaps can be estimated by monitoring the size and shape engineering dimensions with different elements combinations in the QDs.

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2.1.2. Carbon Quantum Dots (CQDs)

It is having small carbon nanoparticles less than 10 nm with some surface passivation modifications. The CQDs having tunable fluorescence emission similar to QDs (inorganic fluorescent) semiconductor. The properties of QDs have a significant and driving role for expansion of CQDs research. Apart from QDs, the CQDs primarily consist of carbon which is non-toxic, which has drawn attractive applications from manufacturing of photonic based devices to biological imaging. The biocompatibility of carbon composites adds extra features to the distinctive and electronics properties of the CQDs which differs from the other nanoparticle families.[37–39]

2.1.3. Gold Nanoparticles

Colloidal systems have been employed to prepare the gold nanoparticles (AuNPs) and, hence called colloid gold. The colloid is basically colored and distinguished based on the diameter of the particles such as intense red for particles < 100 nm or for larger particles which comprised by purple/blue.[40,41] The AuNPs have attracted towards potential applications in various field such as electron microscopy, flexible and stretchable electronics due to their peculiar optical, electronics and biological properties. In ancient times, AuNPs were well known because of their distinctive blue or red color and was widely used by the civilizations in those times. The emergence of this distinctive color is due to the localized surface plasmon resonance (LSPR) in which conduction electrons present on surface of the nanoparticles (oscillates in resonance with the specific wavelength of the incident radiation). Apparently, increment in the diameter of the nanoparticle increases the wavelength of the absorbed light. As an example, gold particles having diameter of 30 nm have peak LSPR absorption at 530 nm. It is important to note that the environmental factors including refractive index of surrounding can change the materials light absorption. Therefore, the surface of nanoparticles or its surrounding solvent, the absorbed molecular interaction on it greatly facilitates due to detection and recognition of incident light absorbance. Due to chemically stable structure and low resistance that offers gold is a good conductor of electricity which is generally utilized as electrodes over conventional metal films because their arrays can provide superior stretch-ability.[42,44,45] Recently, numerous literatures have been published based on the 0D and 1D nanostructures. These articles are compared and listed in table 1.

Table 1. Efficiency comparison of 0D and 1D nanomaterials.

Sr. No.	Materials	Efficiency	Applications	Ref.
1.	C-60 fullerene, PET/ITO, Polyimide, Al/Pyrex	$V_{OC} = 1600$ V, $I_{SC} = 100$ µA, $P_d = 38$ W/m^2	Power source for watch	[46]
2.	Au particles, PMMA, PDMS	$V_{OC} = 1200$ V, $I_{SC} = 2$ mA, P = 1.2 W	Power source for electronic devices	[47]
3.	Fluorinated-ethylene propylene, Ag- nanoparticles	$V_{OC} = 3.6$ V, $I_{SC} = 100$ mA,	Lighting commercial LEDs	[48]
4.	PDMS, $SrTiO_3$ nanoparticles	$V_{OC} = 338$ V, $P_d = 6.47$ W/m^2, Q = 19 nC/cm^2	Lighting commercial LEDs	[49]
5.	$BaTiO_3$ nanoparticles, PVDF, PA6 electrode	$V_{OC} = 900$ V, Q = 34.4 µC/m^2	-	[50]
6.	PVDF, Al – electrode, rGO–Ag nanoparticles	Q = 50 µC/m^2 P = 5.4 mW	Power candidate for electronic devices	[51]
7.	Graphene oxide- Poly(caprolactone)fibers/cellulose paper	$V_{OC} = 120$ V, $J_{SC} = 2.5$ mA/m^2, Pd = 72.5 mW/m^2	Eco-friendly power sources for biomedical devices	[52]
8.	Cellulose nanofibers, Ag nanowires	$V_{OC} = 21$ V, $I_{SC} = 2.5$ µA, $P_d = 0.693$ W/m^2	Energy harvester and sensor	[53]
9.	Eu-PVDF nanofibers	$P_d = 26$ µW/cm^2	Self-powered energy sources	[53]
10.	Cellulose nanofibers, allicin	$V_{OC} = 7.9$ V, $I_{SC} = 5.13$ µA	Self-powered biodegradable TENG	[54]
11.	Nylon nanowires, PVDF	$V_{OC} = 1163$ V, $J_{SC} = 11.5$ µA/cm^2, $P_d = 26.6$ W/m^2	In self-powered UVR detection device	[55]
12.	Polyimide nanofibers	$V_{OC} = 753$ V, $I_{SC} = 10.79$ µA, $P_d = 2.61$ W/m^2	Wearable energy harvester	[56]

2.2. Thin Film Fabrication of 0D Nanomaterials

For the electronics and optoelectronics application, the colloidal NPs are usually fabricated by an efficient and high-

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quality patterned thin-film for device fabrication. Herewith, different fabrication techniques have been employed and thoroughly discussed for NPs colloid solids.

2.2.1. Casting Process

Casting is the simplest technique for film fabrication, in which NP precursor was setup to drizzle on target surface to generate the desired film thickness. The assembly of NP film fabrication is primarily depending on the kinetic deposition process. However, the substrate is frequently adjusted with self-build single layer of organic molecules to confirm the uniform film formation. Casting is mainly divided into two domains namely, spin coating[58] and drop casting.[57]

The drop casting process is a facile method in which the given nanoparticle solution is introduced on the surface and then left for drying. Nanoparticles having long-chain hydrophobic surface ligands namely, oleic acid capped PbSe or InP results into uniform film formation with this technique.[59–61] The drawback of this process is inability to control the thickness and the uniformity of the prepared film because the topography of the surface depends upon the drying procedure adopted and usually edge of the films is thicker than the central part. The major problem in this process to control the desired thickness and uniformity because the surface morphology primarily passing through drying step (the edge of the film is generally thicker than the center). This effect is responsible for the declined output of final devices.

On the other hand, the spin coating[62,63] is known as popular method to fabricate film with desired uniformity, simplicity and reproducibility. In this process, the spin coating machine is used to cast the nanoparticles solution onto the rotating substrate where only the nanoparticle thin film remains because the rotation of the substrate forces the solvent to evaporate into the atmosphere. It is essential to note that the morphology of the thin film in the spin coating process primarily depends upon the rotation speed of the substrate and not on the evaporation of the solvent like drop casting process. Also, the thickness and the roughness of the thin film primarily depends upon the viscosity and the concentration of the nanoparticles. With various advantages of spin-coating processes have been fascinated to the researcher for the fabrication of thin film but it has certain limitations. During the spinning, nearly 10% of NPs solution used to coat on the substrate while other 90% solution get wasted. Also, the spin-coating is popular to develop continuous film unless lithography process has been used. However, to prepare desired film formation some advanced techniques are needed for NP solids such as 2D/3D structures.

2.2.2. Dip Coating

As the name suggests, the substrate is dipped into the jar which contains NP solution. Afterwards, the substrate is drawn out and kept for drying at controlled temperature rate. To ensure the formation of desired thickness film, mainly the speed of drying must be well supervised. The coating thickness cab be theoretically represented by suing Landau-Levich equation:

$$h = \frac{c(\eta u)^{2/3}}{Y^{1/6}(\rho g)^{1/2}} \quad (1)$$

where, η (kg/(ms)), u (m/s), Y (kg/s^2), ρ (kg/m^3), c, and g are represented as viscosity, shrinkage rate, interface tension, solution density, a constant and acceleration gravity respectively.

All the parameters must be regulated with temperature, humidity, vapor pressure and other environmental conditions to get desired thickness of the film. Whereas, determining factor is controlled by shrinkage rate during the process. By adjusting the extraction rate, the film thickness can be controlled from few nm to several 100 nanometers.[24,64,65]

2.2.3. Langmuir–Blodgett Deposition (LBD)

Conventionally, amphiphilic organic molecules were used in the LB deposition technique and is considered as excellent technique for preparation of the self-assembled monolayer from nanoparticles.[66] In this procedure, the deposition of NP as LB layers on the substrate target can be effectively achieved well-defined film with the help of organic ligands on the colloid surface. The pressure which is induced on the surface of the substrate, monitors the shape of the film and it is possible to control it via movement of the sliding barriers on the surface of liquid. It is possible to monitor the surface pressure on the substrate precisely by the injection amount to achieve the uniform nanoparticle layer and vary its assembly density. LB technique has been utilized to obtain various shapes by manipulating NPs such as nanodots, nanorods, nanocubes and nanowires.[67] With enormous advantages of this technique, has been popular for achieving desired shaped assembly of NPs. On the other hand, this technique suffers with various constraints. In this process, the NPs disclosed to water body interface, it prompts serious consequences to the moisture sensitive materials for specific types of materials. Additionally, devices (conducting polymer) and NPs (organic molecules) itself are damaged by water body. The large-scale production is poor due to LB membrane which is continuously floating on the surface of the liquid.

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2.3. Pattering Techniques of 0D Nanomaterials

For high density application, the electronic and optoelectronic device was fabricated using NPs need to be pattering. The evolved patterning techniques allowed us to achieve complex patterns on many substrates and also to fabricate layer-by-layer geometry for the flexible devices. Herewith, different pattering methods have been employed and discussed in details.

2.3.1. Screen Printing

This technique is very economical, effective and, hence under environmental condition it can produce large number of identical patterns in one printing step for NPs. This technique is quite similar to doctor blade, except screen patterned is presented. In this process, the ink is injected through patterns screen to the target substrate and to avoid any leakage of ink, the viscosity should be maintained at specific value.[68] For pattering, the binder popularly used is poly-methyl methacrylate (PMMA) within the NPs homogeneously for better adhesion as collectively on the substrate.

2.3.2. Inkjet Printing

This process quite popular for the fabrication and assembling of different fields such as solar cells, light emitting diode (LEDs), and field-effect transistors (FETs) and used for the interconnection in electronics circuit. Inkjet printing process has been an attractive method recently due to microscale pattering for various types of NPs. The NPs are dispersed with the ink homogenously in a solvent and sprayed thorough tiny droplet with the help of electric field at the preferred location of substrate. Due to microscale pattering, the loss of material is very less and inkjet printing preferred which regarded as inexpensive as other pattering method. Different substrates has been used in this process for pattering such as polymers, metals and biomaterials.[69,70]

2.3.3. Microcontact Printing

This process is very effective for pattering at micro and nanoscale design by utilizing stamp at room temperature. The stamp is made up of soft elastomers through which ink/NPs hit on the target substrate. This technique has been classified into three categories such as additive, subtractive and intaglio transfer. Primarily, additive transfer utilizes spin or spray casting for the coating of NP layer on elastomeric stamp. The homogenous mixture of ink/NPs have been ejected by maintaining the dimensions on to the target substrate.[71,72a]

2.4. Application of 0D Nanomaterials

As a result of different thin film coatings and pattering techniques are explored in the above sections have been immersed purposely and used to fabricate diverse nanomaterials. By utilizing these techniques, the range of nanomaterials from metal NWs to QDs have been procured and thoroughly used in electronics and optoelectronics devices. These nanomaterials can be served in different devices as conducting layers/electrodes, semiconductor channels, electron/hole pair transporting layers, light emitting/light absorbing layers etc. The specific applications are elaborated of the assembled NPs in the coming section.

The simplified structure with higher output performance, the surface modification of nanoparticle has enabled and reported by Zhou et al.[72b] The gold (Au) nanoparticles procured by reducing sodium tetrachloroaurate (III) dihydrate onto the surface of Au film assembled uniformly and observed by SEM micrograph [Figure 2(a)]. In between the PDMS thin films and nanoparticle decorated Au thin film, the contact friction occurs. The four springs are positioned at the corners which create a small distance between the contact materials and promotes the contact-separation cycles with external mechanical stimulus. The current output performance of TENGs with and without Au nanoparticle decorations [Figure 2(b–c)]. The maximum peak current of TENG with Au nanoparticles can reach up to 150 mA with a 10 N of contacting force (~4 times higher than TENGs with a flat Au contacting layer) which in turn gives a 25-fold power output improvement. Aforesaid, the excellent efficiency was achieved because of nanoparticle-modified TENG (enormous contact area provide by the decorated Au nanoparticles) and large deformation of PDMS that leads to conformal contact despite the roughness of the surface.

Further, Liu et al.[72c] introduced a thin film blue energy harvester which is primarily based on liquid-solid contact TENG and the structure of final device is maintained as flexible and bendable at the substrate. Primarily, by minimizing shielding and maximizing output performance, the external U electrode (including a bar electrode, B electrode) and a U-shape electrode (U electrode) is introduced to fabricate the device. The final TENG device is utilized for seashore IoT system application, where U electrode can draw electric charges from flood's wave and B electrode act as safety switch. The whole device system can sense wave warning system (temperature sensor) and continuous power supply to the normal operation. Eventually, the U electrode comprises with wireless transmitter which successfully send the information to the cell phone. For assembling the device, the U-TENG is developed from the dielectric electrode TENG (D-TENG), and D-TENG comprises to the single-electrode mode of four basic working modes.[72c,d] The different structure is depicted in

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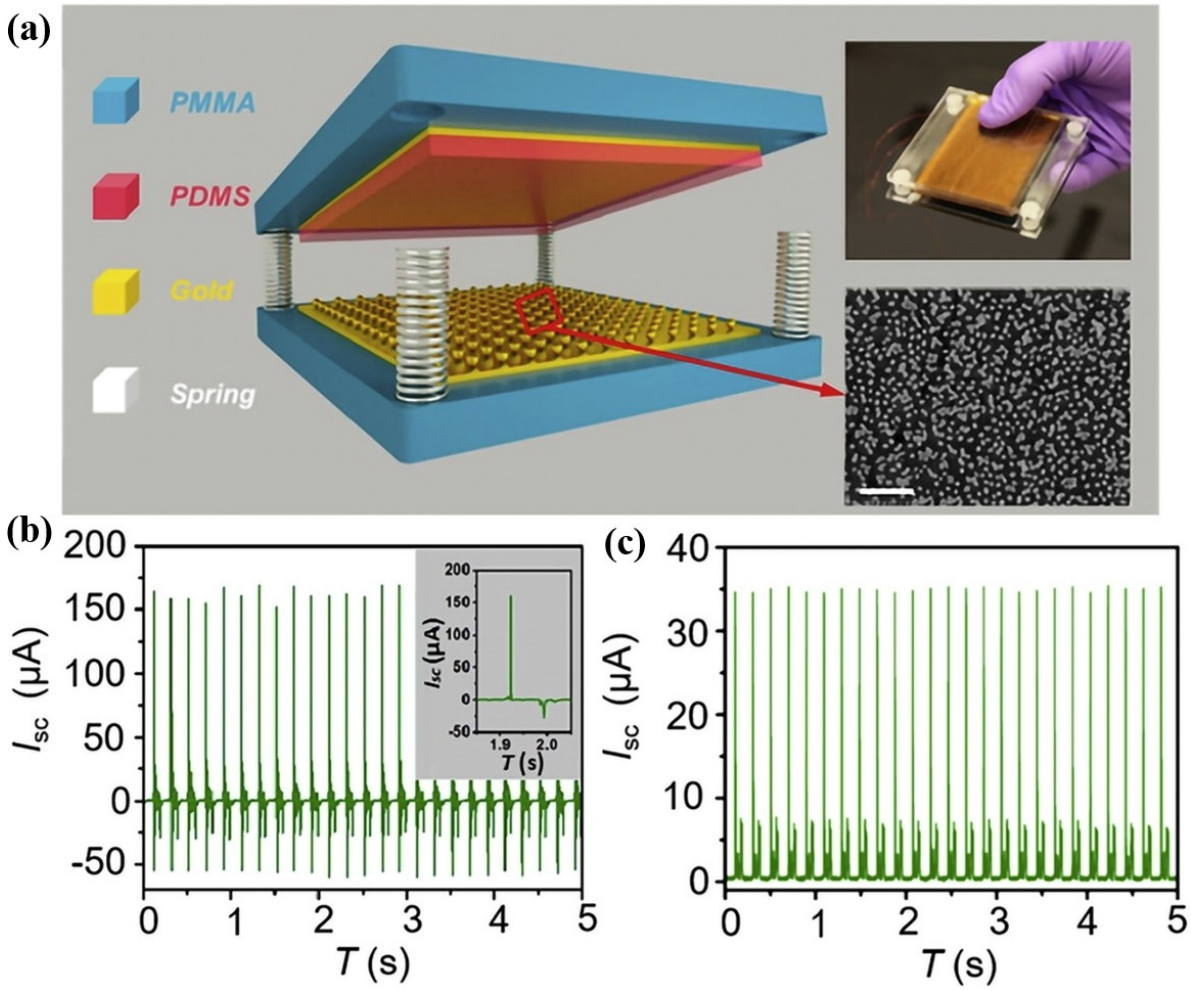


Figure 2. (a) Au-nanoparticle-based TENG (Scheme, optical image, and SEM, Scale bar, 500 nm), and (b)–(c) *Isc* of the TENG with and without Au-nanoparticle decoration..[72b] [Adapted with permission from ref. 72b Copyright 2020, Elsevier]

Figure 3(a–c), where the working principle and efficiency outputs are provided for D-TENG and U-TENG. Firstly, the electrical double layer (EDL) formed on the interface between the film and water waves; when water hit the surface of dielectric film the surface group may lead its surface to be negative charged and EDL layer will form to neutralize these charges.[73a] The EDL layer consists of a Stern and diffuse layer, where Stern layer refers to ions adsorbed on the dielectric surface and diffuse layer contains mobile ions which are loosely bound with solid and can easily move with water wave. The ions present in the diffuse layer are moveable at large scale which is heavily screened at surface of dielectric and outputs of liquid-solid contact mode TENG are improved.[73b] As shown in Figure 3(b), the working mechanism of D-TENG is classified into four steps: (i) the dielectric film is pre-charged as the new cycle starts with negative charges because of preliminary action of water waves, and loosely positive (+Ve) and negative (-Ve) ions are distributed in water arena (ii) the EDL layer formed on the interface when water wave forms a contact with the dielectric film. Wherein +Ve ions are attracted to form the first layer and loosely -Ve ions displaced with water to form the second layer. In general, the equilibrium of electrical neutrality at initial state is allocated, and electrical signals are created in the external circuit and passes to ground (iii) the EDL is steadily immerse the dielectric film along with the distributed equilibrium until it is completely covered which form new equilibrium of electrical neutrality (iv) whereas, the water wave breaks off and separated from dielectric film, the EDL influence recedes and electrical charge is created in external circuit and pass to the dielectric electrode. When, the water wave is entirely away which leads to its original position as state (i). This working phenomenon

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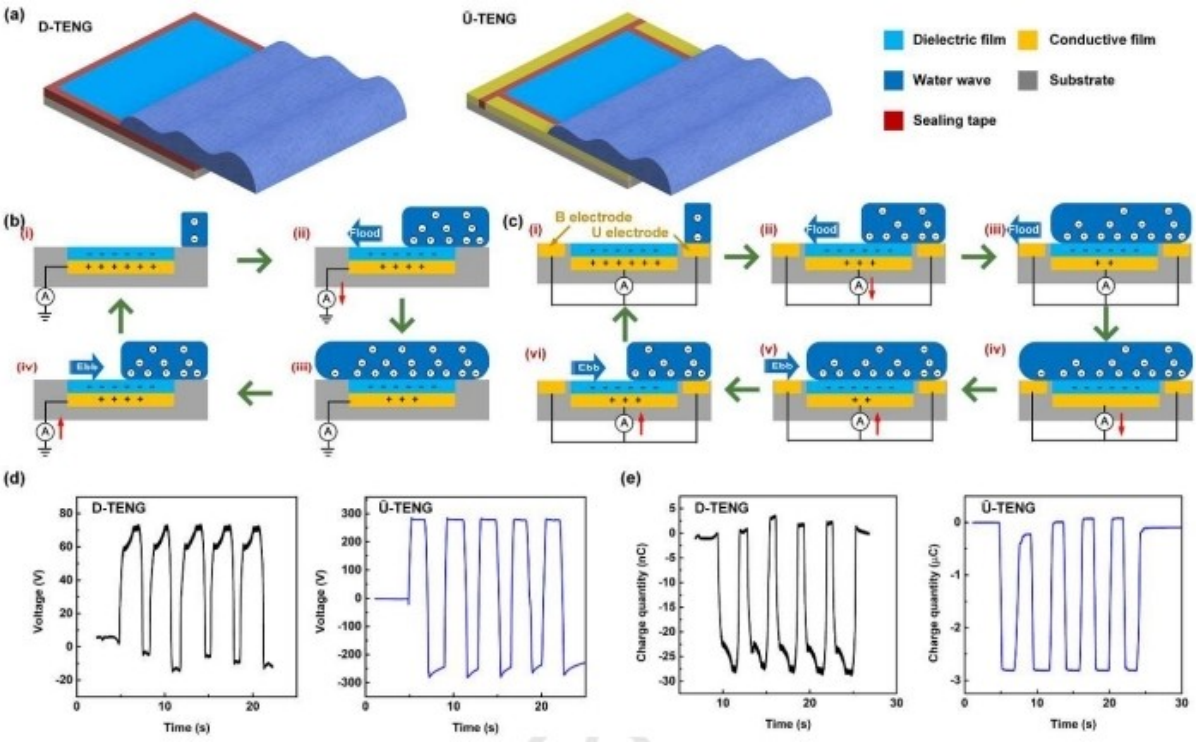


Figure 3. (a) The structure of the D-TENG and the Ū-TENG, (b)–(c) Working phenomenon of D-TENG & Ū-TENG, (d) Voc comparison of the D-TENG & Ū-TENG, and (e) Comparison of transfer charge quantity of D-TENG & Ū-TENG.[72c] [Adapted with permission from ref. 72c Copyright 2019, Elsevier]

of D-TENG attributed to poor charge flow in the external circuit due to shielding effect from the water which resulted lower output efficiency. Also, the loosely -Ve ions in the second layer screen the charges on the dielectric surface that will limit the output of TENG.

Further, the U-TENG exploit external U electrode/U-shape electrode to minimize this shielding effect from the second layer, where mechanism is thoroughly explained [Figure 3(c)] in six steps: (i) as the cycle begins with the dielectric film is pre-charged with -Ve charges because of preliminarily friction from the water waves, and loosely +Ve and -Ve ions are uniformly distributed in water arena (ii) once the water wave touches with U electrode, the loosely ions are affected due to potential difference between the D electrode and U electrode, and the negative ions are inclined to retained in the water area from the D electrode. The EDL layer formed to maintain the electrical neutrality for the liquid-solid contact area. Finally, with the motion of the water wave, the initial equilibrium is constantly distributed which causes to flow of electrical current from D electrode to U electrode (iii) the temporarily equilibrium is obtained between the D electrode and U electrode when water wave entirely covers the area of these electrode. At the liquid-solid contact area interface, the EDL is formed, there may still exist loosely negative ions in the second layer (iv) the water wave contacts with B electrode when water wave across the D and U electrode, remained loosely negative in the second layer which further can attracted towards the B electrode because of electric potential difference from D electrode, thereby electrical signals are generated in the external circuit which flow from D to B electrode (v) after the flood process, the water wave begins ebb away from the device where loosely ions absorbed on B electrode will get away with the water wave. Therefore, the equilibrium of electrical neutrality is broken which causes the generation of electrical signals in the external circuit flow from B electrode to D electrode (vi) lastly, the water wave begins ebb away from area of D electrode where the equilibrium is continuously broken, and electrical signals in external circuit flow from U to D electrode until water wave completely away from the device. Subsequently, the cycle is back to the initial state of (i). In comparison with D-TENG and U-TENG, the U-TENG is little concerned with the shielding effect from the water due to U and B electrode attracted loosely ions and exclude their screen affect the charges in dielectric film. The comparison of output efficiency is depicted in Figure 3(d–e). The Voc enhances from 73.1 V to 283.1 V, transfer charge quantity increases from 28.4 nC to 2.8 μC and short-circuit current increases from 0.4 μA to 10.8 μA. Altogether, the U electrode

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along with B electrode and U electrode minimize the shielding effect which enhance the outputs. Also, it adds another benefit of the U electrode about dynamic change of the water film on liquid-solid interface in the ebb process.[73c–d]

Furthermore, the 0D applications are widely explored in different devices as conducting layers or electrodes, light emitting diodes and transistors etc. In this section, the specific applications are elaborated of the assembled NPs.

2.4.1. Electrodes

The metal NPs which are dispersed in ink designated as colloidal metallic has been extensively used to produce electrodes. The colloidal process of NP solutions having numerous benefits than conventional vacuum evaporation process particularly low processing with low instrument cost, mild operating requirement, and conformity with electronic system. The inkjet printing has been utilized to make a conductive ink which is composed of Ag NP concentrated solution. With monitoring the proper ratio of NPs, surfactants, stabilizers, solvents and different pattering method by controlling electrical properties, the electrodes can be formed at various substrates. However, the Ag grid has been reported by Kreb's group by utilizing roll to roll printing of Ag NP inks.[73e] This electrode is potentially utilized for sensor applications and it has been also used in unconventional microelectronic for interconnections. The interconnecting electrodes has been vastly used in 3D antenna structure, and for electronic circuit.[74]

2.4.2. Light Emitting Diode (LEDs)

LEDs are of primary importance in the field of optoelectronics because of their high luminosity and brightness. Substantial amount of research has been done on colloidal semiconductor nanoparticles as device components in LEDs, like in color filter, emission layer and the electron transport layer.[75] Commercially in the present scenario, the application of QDs film in backlight units in order to achieve high color purity in displays is gaining wide attention. Backlight units which are primarily based on conventional white LEDs can only provide a color gamut of less than 70% of the standard provided by NTSC. In order to achieve the natural and vivid colors, inclusion of QDs was performed in the backlight unit of the displays. The fabrication of colorfilters and converters is achieved by mixing the nanoparticles with polymers and casting them on the blue LEDs.[76] The blue light from the blue LED chip is down-converted to a longer wavelength when it transmits through nanoparticle embedded polymer film. It is possible to achieve the reproducible and tunable colors from the color filters because of unique properties of QD namely, narrow emission spectra, high luminescence efficiency and facile color tunability.[77] Visuals of the active-matrix LEDs with colors filters are provided in Figure 4(a–d). Currently, many industries have been extensively using QD layers in their electronics displays. As a commercial application, QD Vision has fabricated QDs based color filters and implemented them in liquid crystal display where they successfully achieved color gamut of nearly 100% of the set standard.

2.4.3. Transistors

The next generations NPs based semiconductor have been emerging components for electronics devices. The NP FET has been thoroughly used as essential component in integrated circuits. It is possible to realize several devices namely, logic gates, amplifiers and inverters with the merging of n- and p-type transistors. Addition for colloidal nanoparticles in FETs have several important advantages such as their compatibility with flexible substrates, solution processability over the large areas and low cost of the processes. Numerous types of NPs semiconductor such as CdSe and PbS have been employed in transistors as semiconducting channel. Altogether, in the fabrication of transistors, the NPs are spin casted on the oxidized Si wafers where the thin SiO_2 layer act as a dielectric layer and the Si wafer will act as a back gate. To enhance the charge carrier mobility, oleic acid ligands with hydrazine has been researched. Kagan et al. recently reported flexible NP based transistors,[79] a solution processed CdSe transistor has been fabricated with the help of Kapton film [Figure 5(a)].[80] To avoid the leakage current of the assembly, thiocyanate-exchanged CdSe NPs was spin coated on ultrathin Al_2O_3 dielectric layer [Figure 5(b)]. With this assembly, the high-performance was achieved as low hysteresis, low-voltage CdSe transistors with 22 cm^2/Vs of electron mobilities [Figure 5(c)]. Likewise, various aligned nanowires (NWs) have been utilized for flexible transistors.[81] For example, PbSe NWs was culminated with hexadecane-graft-poly(vinylpyrrolidone) co-polymer and drop casted on Al_2O_3 and octadecyl phosphonic acid (double dielectric layers) under the influence of external electrical field across the device structure [Figure 5(d)]. The electrically aligned PbSe NWs SEM micrographs demonstrated where I_D (A)–V_G (V) curves have shown reduced hysteresis and transistor operation at lower voltage [Figure 5(e–f)].

2.5. 1D Nanomaterials

For the application of soft electronics from electrodes, conducting wires, sensors, and energy devices, 1D- nano-materials have been extensively explored in recent years. For flexible and stretchable devices, various materials with hetero-structures have widely explored such as carbon nanotubes (CNTs), metallic nanowires (NWs), metal oxide NWs,

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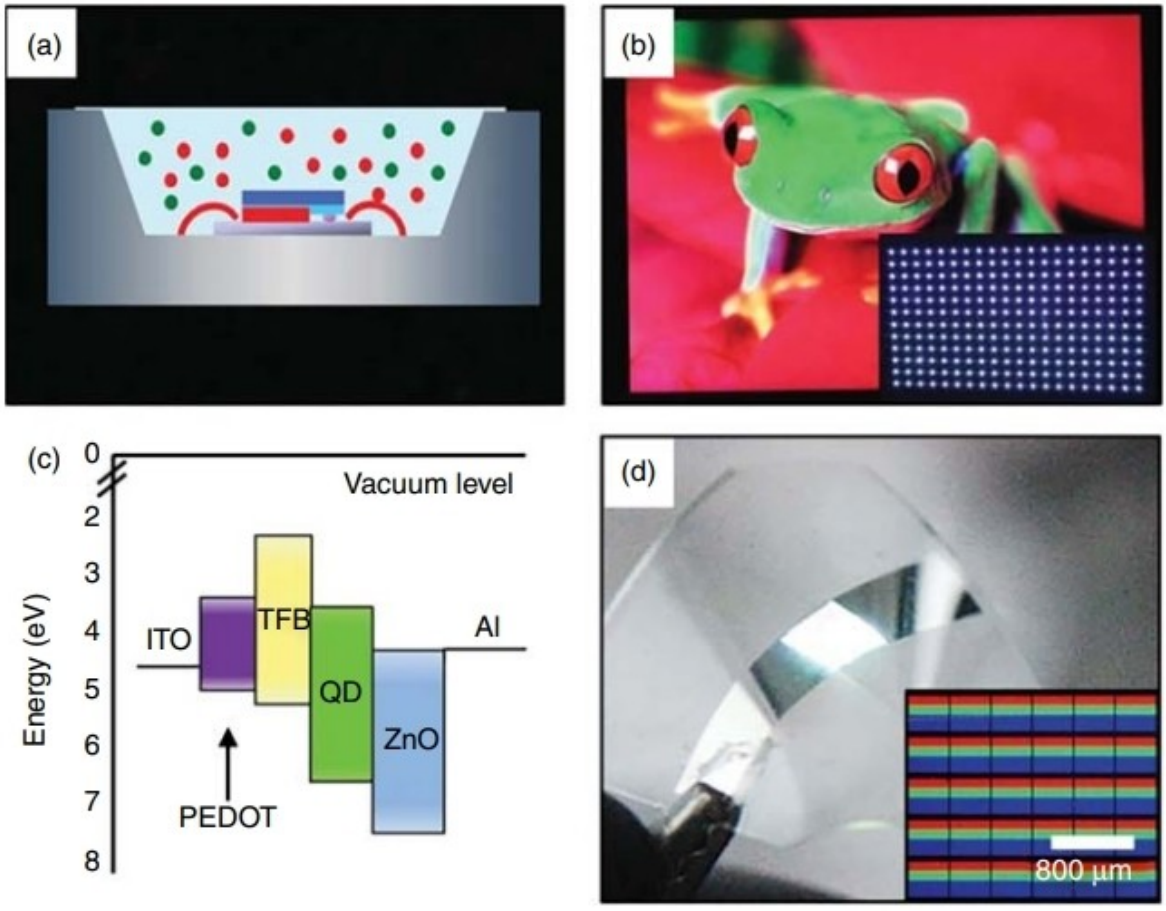


Figure 4. (a–d) LEDs using colloidal NCs, (a) Side view of QLED; green and red quantum dots are captured inside a silicone resin and used as color converter for blue LEDs in order to produce white light, (b) Visuals of 46-inch LCD TV panel using white QLEDs as provided in (a), The inset displays the white QLED backlights, (c) Band diagram of intaglio transfer-printing process fabricated QLEDs and energy level calculations were done by using UV-photoelectron spectroscopy, and (d) Optical image of operating flexible white QLEDs fabricated using the same intaglio transfer printing process. Inset: magnified view of RGB QD pixels.[78a,b] [Adapted with permission from ref. 78a,b Copyright 2010, John Wiley & Sons]

polymer NWs, and nanofibers as 1D nanomaterials. One-dimensional nanomaterials with high aspect ratio that forms high-performance electrical and mechanical behavior. Various types of 1D material with their synthesis and application for TENG have been discussed.

2.5.1. Carbon Nanotubes (CNT)

2.5.1.1. Fabrication Method of CNTs

For the current generation, the CNTs have been exercised for the flexible and stretchable electronics. The CNTs have been fabricated by two techniques such as bulk materials (CNT-elastomer composite) and surface materials (stretchable materials).

2.5.1.2. CNT-Based Bulk Materials

The fabrication of CNT-elastomer based composite generally fabricated with the dispersion of CNTs into polymer matrix. In this succession, Sekitani et al. procured composite film by dispersing single wall carbon nanotube (SWCNT) into polymer matrix i, e vinylidene fluoride-hexafluoropropylene copolymer with the help of ionic liquid 1-butyl-3-methylimidazolium bis(trifluoromethanesulfonyl)imide.[83] Again, Majid et al. reported dispersion of polydimethylsiloxane (PDMS) into toluene and formed composite of CNT-PDMS membranes.[84] The main purpose of using toluene solvent media, where CNTs easily dispersed with higher degree of uniformity. By tuning the dimensions of CNT-elastomer composites such as micro and nanostructures, the performance can be enhanced which can widen the scope of the application in the various field. Further, Jung et al. fabricated wearable

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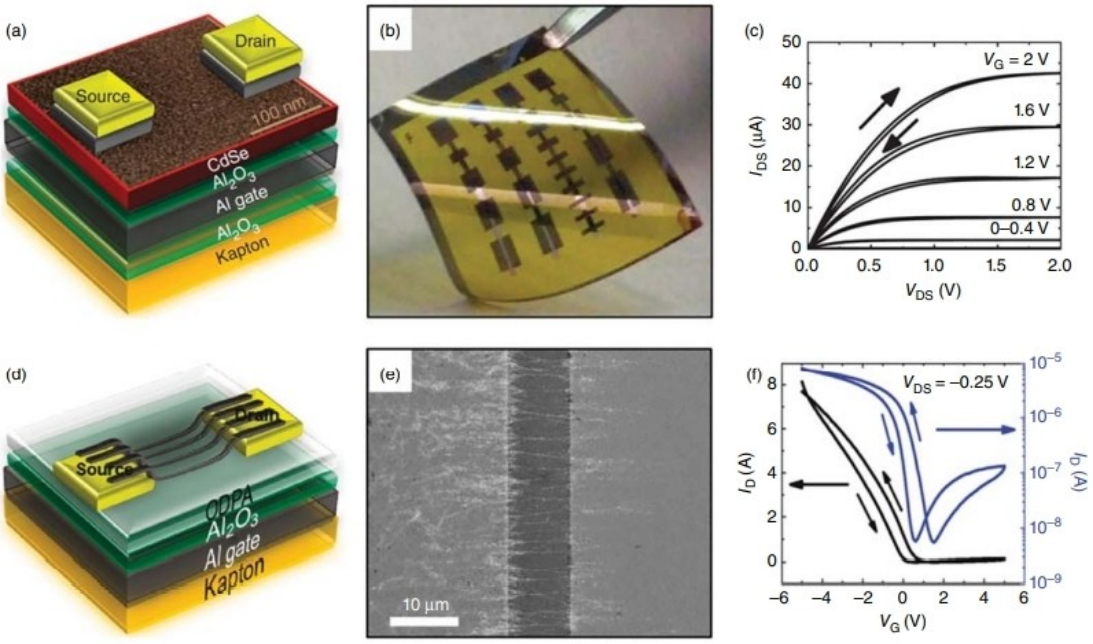


Figure 5. Transistors using NP layers (a) A schematic of the device structure of a flexible CdSe NP field-effect transistor, (b) A photograph of a flexible CdSe NP field-effect transistor, (c) Output drain current versus drain–source voltage characteristics of the device,[80] [Adapted with permission from ref. 80 Copyright 2012, Springer Nature] (d) Schematic of a flexible PbSe NW field-effect transistor,[81] [Adapted with permission from ref. 81 Copyright 2011, American Chemical Society] (e) An SEM image of PbSe NWs aligned on a gold source–drain electrode,[82] [Adapted with permission from ref. 82 Copyright 2013, Royal Society of Chemistry]and (f) I D–VG characteristics showing the reduced hysteresis and low-voltage operation of PbSe NW transistor[81] [Adapted with permission from ref. 81 Copyright 2011, American Chemical Society]

human-machine interfaces by using reverse-micelle-induced porous pressure-sensitive rubber. The complete dispersion of CNT and reverse micelle solution (RMS) into the PDMS solvent, later which can frame porous structure.[85] Meanwhile, Ma et al. prepared CNT/PDMS composites by utilizing microelectromechanical system (MEMS) technology. In this process, the lithography technique was utilized to propose SU-8 photoresist mold for assembling the sensor array with nylon membrane filter. Through this array, transferring of micro domes onto the conductive polymer film, which can improve the sensitivity of the CNT/PDMS composites for tactile sensors application.[86]

2.5.1.3. CNT-Based Surface Materials

In this technique, the random allocated CNT directly deposit on the surface of flexible and stretchable substrates by using different techniques such as spin coating, chemical vapor deposition (CVD), and transferring of freestanding CNT film. In this continuation, Lipomi et al. procured nanotube films by utilizing spray coating onto ultraviolet/O_3 surface with PDMS. The estimated value of square resistance ~328 ohm/sq, transmittance ~79% at strain of 0%. Furthermore, pre-stretching PDMS can solve the issue of irrecoverable loss of junction between the nanotubes which minimizing the resistance which was caused with increased strain.[87] In similar fashion, Chen et al. stated uniaxially pre-stretched PDMS, well-aligned CNTs, deposition of polyaniline (PANI) and then after ordered wavy CNT might be accomplished.[88] The free-standing array of CNT film network can be achieved after transferring to PDMS substrate with higher transparency illustrated by utilizing floating catalyst chemical vapor deposition (CVD).[89]

2.5.1.4. Application of CNT

The CNTs having unique advantages and types of fabrication techniques can be extensively utilized in various applications such as wearable/stretchable sensor, energy harvesting, electronic skin method etc. Yamada et al. fabricated thin film which is aligned with CNTs and further utilized as wearable and stretchable devices in Figure 6(a). When this fabrication alignment was stretched, the nanotube films get fracture inside the gaps, islands, and bundles, gap bridging. This mechanism process leads to strain phenomenon sensors which was proficient to estimate strain about 280% with lower creep, higher durability and rapid response (50 times extra strain than common metal strain gauge).[90] It has been utilized the

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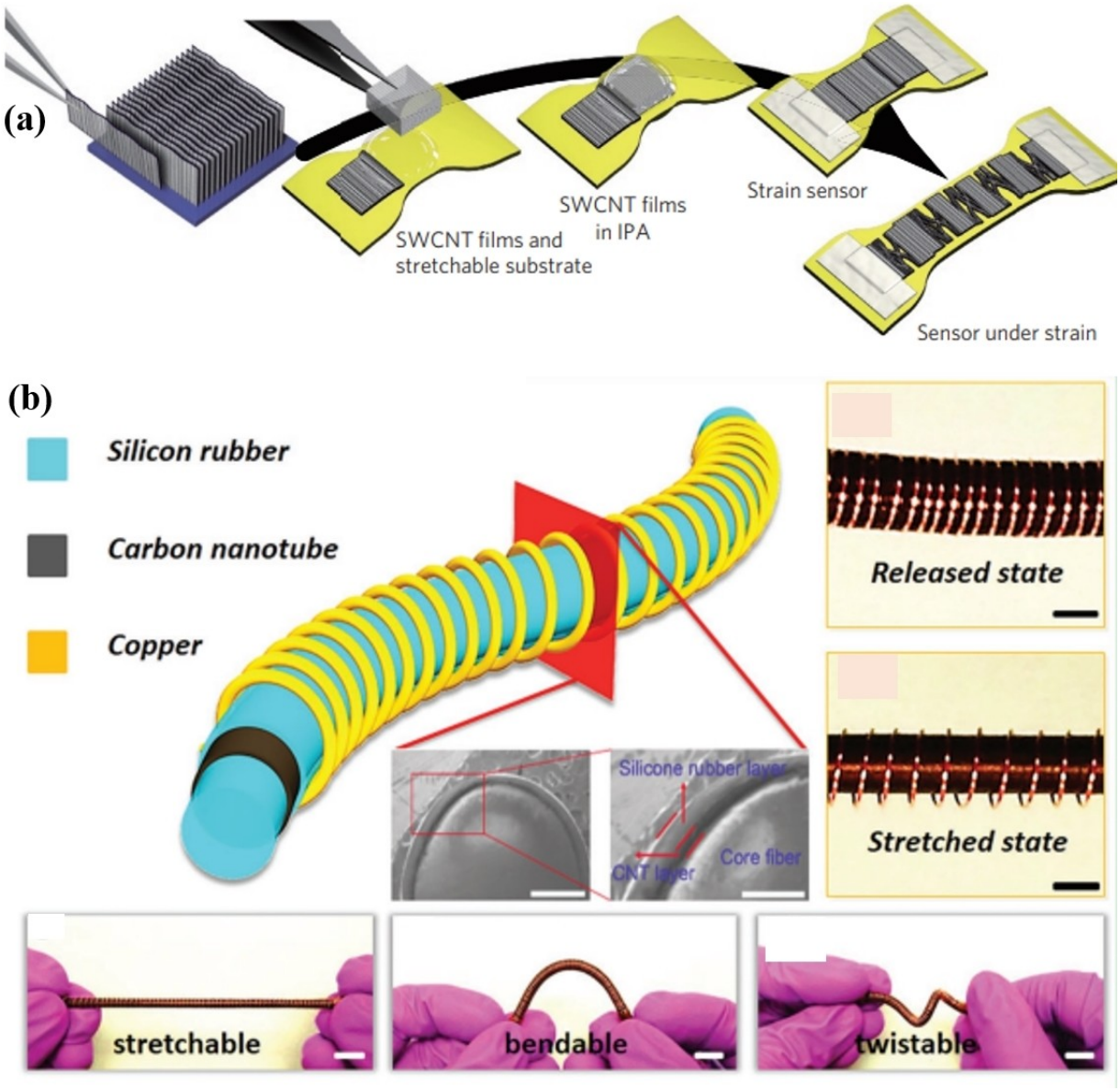


Figure 6. Application of carbon nanotubes (a) Schematics of fabrication of the single-walled carbon nanotubes strain sensor,[90] [Adapted with permission from ref. 90 Copyright 2011, Springer Nature] and (b) The structure of the fiber triboelectric nanogenerator in a bent state, stretched state, twisted state and released state, scale bar: 6 mm[91] [Adapted with permission from ref. 91 Copyright 2016, John Wiley & Sons]

CNT based sensors into the stockings, bandages, and gloves which can identify the various human motion movement including typing, breathing and talking. In this continuation, for power generation Xu et al. have developed novel based TENG utilizing high-stretchable and nanofibers. During fabrication, the CNT was coated on Si rubber fiber which is utilized as stretchable electrode, composite of ink/CNT and stretchable polymer was consequently assembled as fiber based TENG with the coating phenomenon [Figure 6(b)].[91] To maintain the controllability, the CNT was exposed to the PDMS which can be utilized as +Ve triboelectric layer, hence for TENG pairing with polymeric materials like polythene, terephthalate (PET), polyimide (PI), PDMS, and polytetrafluoroethylene (PTFE) could be done. The estimated output voltage, and current density of CNT-PDMS TENG was achieved 150 V and 60 mA/m^2 which is more significant (250% and 300% of its value) than PDMS/MWCNTs nanotubes.[3d]

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2.5.2. ZnO Nanowires (ZnO NWs)

ZnO is popularly known as semiconductor material with wide band-gap can be utilized for energy harvesting system such as piezoelectric properties. This is called "self-powered nano systems" when piezoelectricity textile fibers coated with ZnO nanomaterials and has been capable to harvest the energy from movement of wind and body. Also, ZnO can be used thoroughly for biomedical applications due its non-toxicity properties. Based on the nanostructures of ZnO (NWs, nanobelts, nano springs, nano rings, nano bows, and nano helices etc.) and their application can be diverse for potential fields. For self-powered system, ZnO NWs play a crucial role as energy harvesting system.[92] In this succession, Wang et al. reported nanoscale mechanical energy to electrical energy via piezoelectric arrays of ZnO NWs. The aligned NWs are deviated by atomic force microscope (AFM) which is conducting tip contact mode. Bending of coupled piezoelectric and semiconducting system of ZnO, it generates a strain and charge polarization throughout the NWs.

The ZnO NWs results for electrical generation, when Schottky barrier established between metal and NW. The performance of NW-based piezoelectric power generator is optimized about 17 to 30%. Several advances of ZnO NWs resulted potentially for converting mechanical, vibrational and hydraulic energy into electricity for powering nanodevices. The basic characteristics of generating electricity is the deviating the crystal structure symmetricity under the influence of external force mechanically which forms piezo potential Figure 7.[93]

2.5.2.1. Preparation of ZnO Nanowires

To prepare the high aspect ratio ZnO NWs by optimizing morphologies such as aligned NWs, higher surface area, the different approaches have been defined for potential applications. Most favorable synthesis method to prepare ZnO NWs is vapor-liquid-solid (VLS)[94] by maintaining temperature of 90 °C with aqueous solution of zinc nitrate ($Zn(NO_3)_2$) and hexamine ($C_6H_{12}N_4$) in basic environment. To enhance the aspect ratio of ZnO NWs, some additives like polyethylene glycol or polyethyleneimine can be added.[95] Cui et al. prepared the ZnO NWs by using different metal salts other than zinc nitrate with growth solution approach.[96] During synthesis, Greene et al. reported different substrates (preceded silicon, glass and gallium nitride) have been used for the growth zinc salts (zinc nitrate, zinc acetate) within the basic environment.[97] Although, pre-seeding substrate generates sites for homogenous nucleation of ZnO crystal structure during preparation. Further, Wu et al. reported in-situ thermal deposition of zinc acetate crystallites with pre-seeding methods, ZnO NP with spin coating, and utilization of physical vapor deposition (PVD) technique to deposit thin films of ZnO.[98] Top-down pattering method in conjunction with pre-seeding such as electron beam and nanosphere lithography to identify nucleation sites aforementioned growth. Also, Hua et al. reported the aligned ZnO NWs has been utilized for dye-sensitized solar cells and field emission devices.[99] Pu et al. fabricated flexible based TENG by utilizing PVDF/nylon electrospun nanofibers with the alignment of ZnO NWs.[100]

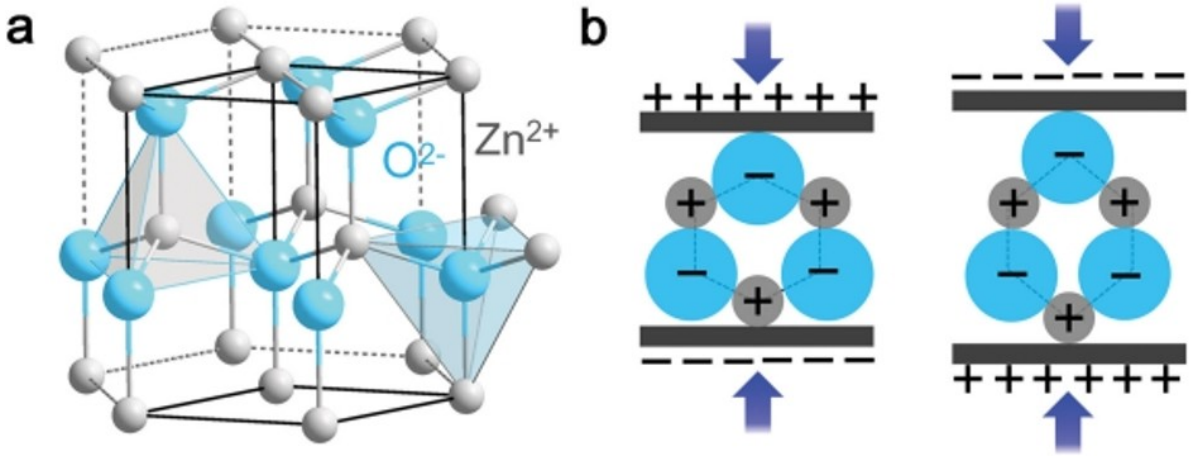


Figure 7. Schematics of fundamental principle behind the piezoelectric effect (a) visuals of the wurtzite-structured ZnO, and (b) Piezoelectric properties and induced polarization in tensile and compression modes of the material[93] [Adapted with permission from ref. 93 Copyright 2016, John Wiley & Sons]

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2.5.2.2. Application of ZnO Nanowires

Due to its semiconducting properties, the ZnO NWs can be utilized as flexible sensors and piezoelectric flexible energy harvesters. Recently, the researchers have explored more possibilities of ZnO NWs for potential application as self-powered system. In this progression, Xu et al. stated that the integrations of vertical and lateral ZnO NWs arrays have been utilized for producing power to function the devices [Figure 8(a)].[101] The ZnO NWs lateral integration generates V_{peak} of 1.26 V at minimum strain of 0.19% which is having capability to recharge AA batteries. However, even single ZnO NW having ability to convert the biomechanical energy to electricity with movement of muscle-driven nanogenerator (NG). Rusen et al. demonstrated the single-wire NG comprises flexible substrate and ZnO NW coupled with laterally at its two ends on the surface of the substrate [Figure 8(b)].[102] While movement of muscle, the NW and substrate stretches back-and-forth. During this process, the piezoelectric potential is created which assist the flow of electrons within the external circuit. The high output voltage optimization is necessary for flexible NG now days for realistic application. To get higher output voltage, Wang et al.[93] integrated numbers of NWs on a single flexible platform to overcome this issue. However, lateral NW arrays on flexible NG integrated nearly 700 rows, ~20, 000 lateral ZnO NWs within each row can be realized. The mechanical deformation was balanced because of the adequate flexibility of substrate which is coupled with NWs array. With periodic mechanical deformation, the maximum voltage was estimated 1.26 V with maximum current of 28.8 nA [Figure 8(c)].

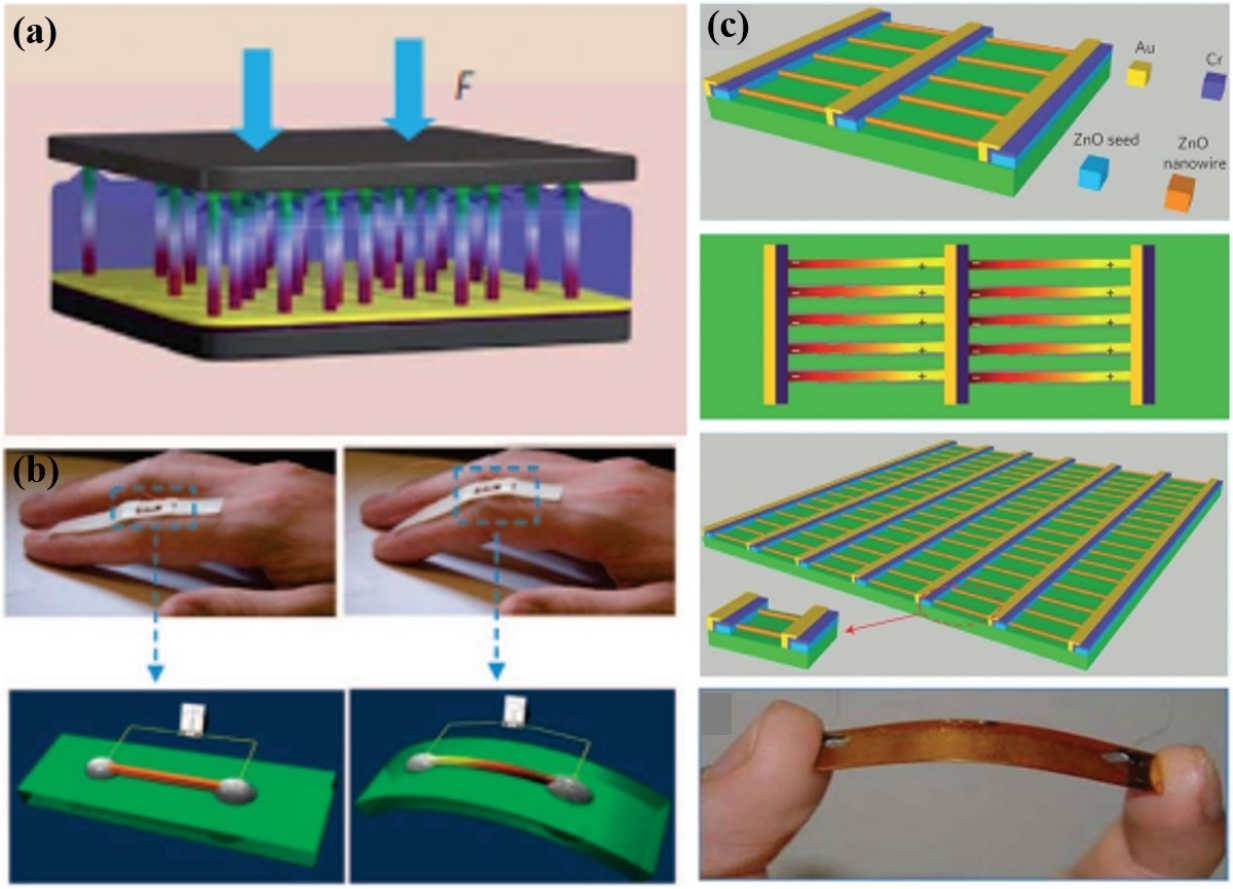


Figure 8. Vertically grown ZnO nanowires (a) Visuals of compressing vertically grown ZnO nanowires and generation of piezoelectric potential in them due to the development of crystallographic strain in the wires,[101] [Adapted with permission from ref. 101 Copyright 2010, Springer Nature] (b) Single wire generator attached to oscillating finger for energy harvesting purpose,[102] [Adapted with permission from ref. 102 Copyright 2009, American Chemical Society] and (c) Structural images of flexible lateral ZnO nanowire array integrated nanogenerator.[93] [Adapted with permission from ref. 93 Copyright 2016, John Wiley & Sons].

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2.5.3. Ag Nanowires (Ag NWs)

2.5.3.1. Fabrication of Ag NWs

The element Ag having higher conductivity than other metal materials. Although, the Ag NWs are most favorable flexible and stretchable electronic device due to lower resistance. To meet the practical application, the most favorable method to fabricate the devices via direct deposition of Ag NW film on the stretchable substrates that concludes Meyer Road coating, membrane filter, and drop drying. The Ag NW/elastomer was formed by homogeneous mixture of Ag NW and polymer matrix to fabricate highly efficient flexible and stretchable electronics (polymer penetration, Ag NW/polymer solution curing, and 3D framework guiding). For scalable roll to roll deposition of Ag NW network, Hua et al. utilized Meyer-rod-coating approach. The estimated sheet resistance was about 8 ohm/sq with diffusive transmittance of nearly 80 %.[103] The solution filtration is best possible choice for to develop conductive film by utilizing longer AgNWs. By keeping this process in mind, Lee et al. displayed longer Ag NWs in ethanol which was collected on Teflon filter to procured consistent film of the percolation network of AgNW.[104] Also, Zhu et al. reported the Ag NW network was formed by using PDMS solution which was spilled on the Ag NW film, the liquid penetrates into the Ag NW/PDMS composites (lower viscosity and lower surface energy).[105] Further, Gao et al. reported the ice template technique to convert directly monodispersed Ag NWs into the freestanding macroscopic 3D cellular-like architectures without utilizing any cross-linker or any functionalization agent for the very first time.[106]

2.5.3.2. Application of Ag NWs

The conducting and greater stability under deformation (stretched and bended) of Ag NWs broadly utilized for transparent and soft electronics. Ge at al. prepared stretchable sensor based on Ag NWs using low-cost electronic fabric which evaluate mechanical stresses convinced by normal pressure, lateral strain and flexion. The intertwined composite fibers (sensors electrodes) along with piezoresistive rubber (as shell-sensing element) and stretchable/highly conducive Ag NW network utilized to fabricate core electrode. In this electrode, the mechanical sensors component attached to each cross-section between the sensor electrodes. Under the influence of stretching and prolonged usage, the electric fabric presents great show in Figure 9(a).[107] Zhu et al. reported PDMS liquid penetration and fabricated higher conducting and stretchable electrode with Ag NWs encapsulated on the surface of PDMS layer [Figure 9(b–c)].[105] Choi et al. represented the heating element (soft and stretchable, light-

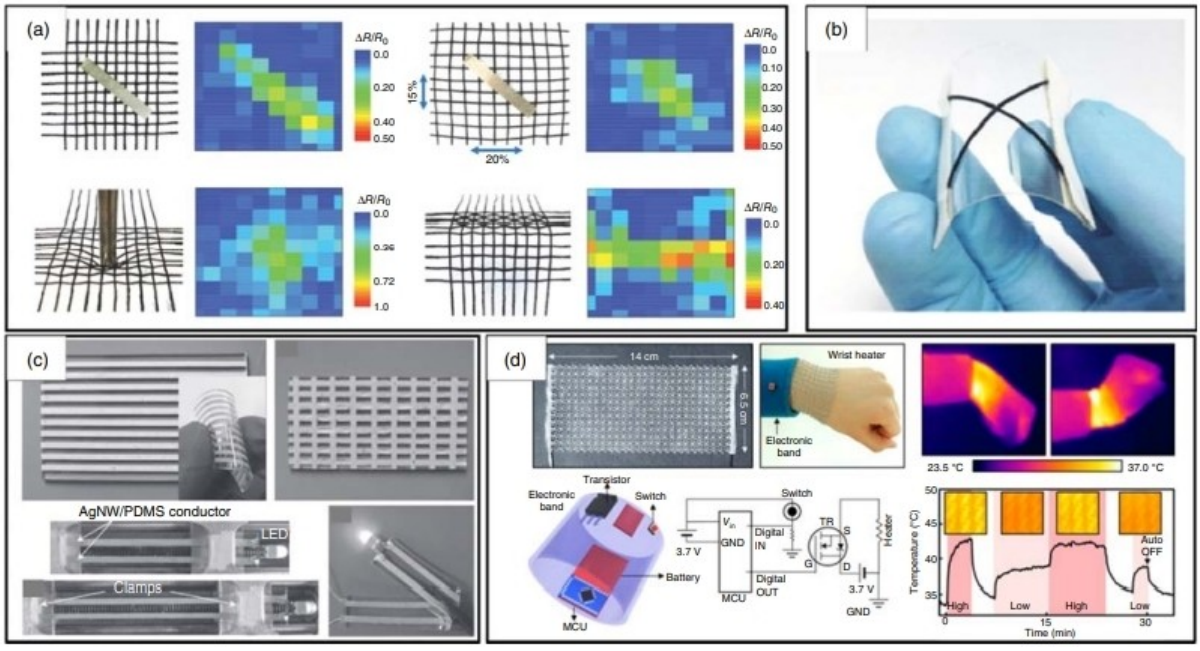


Figure 9. Ag nanowires applications (a) Visuals of force mapping of the electronic fabric with stretchable sensor arrays,[107] (b) Mechanical–electric properties of the sensor unit,[107] [Adapted with permission from ref. 107 Copyright 2015, John Wiley & Sons] (c) Optical images of LED integrated circuit when placed under tensile strain of 0 % and 50 %; also the LED integrated circuit during the operation of folding,[105] [Adapted with permission from ref. 105 Copyright 2012, John Wiley & Sons] and (d) Elastic heater fabrication with a ligand-exchanged Ag-nanowire nanocomposite for wearable articular thermotherapy[108] [Adapted with permission from ref. 108 Copyright 2015, American Chemical Society]

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weight and thin) which is integrated with human joint and skin for efficient heat transfer and thermotherapy. The heater is composed of AgNW/elastomer nanocomposites which is highly conducting for electron charge transfer [Figure 9(d)].[108]

2.6. 2D and 3D Nanomaterials

2D materials have drawn lot of attention over the last decades owing to its electrical properties, mechanical, flexibility, transparency, and a large specific surface area. Furthermore, it is exceptionally flexible and mechanically robust, as well as its higher electrical/ionic conductivities favors for ultrathin-flexible energy harvesting and storage solutions. Based on the material structures, the TENG can be differentiated into three categories: (i) 2D material as TENG electrode, (ii) triboelectric material and (iii) both triboelectric and electrode materials.[109] On the other hand, the 3D materials having several advantages over 1D and 2D counterparts like structural complexity, tunability etc. In recent years, such characteristics are beneficial for flexible and stretchable device which has drawn attention among researchers to explore it extensively. Moreover, it can also provide better interaction of environmental factors to physicochemical reactions. The interaction between mechanics and structures opens up a world of possibilities to design and manufacture unusual 3D structures with innovative functionality. Primarily, the detailed explanation and classifications (Figure 10) of 2D materials have been discussed for TENG applications.

2.6.1. Different 2D Materials

2.6.1.1. Graphene and Graphene Oxide (GO)

Graphene is the most common 2D material explored thoroughly for TENG application due to its high electrical conductivity, surface to volume ratio, flexible nature etc. For TENG device, the high power is required within a small volume of material where graphene is significant candidate in this regard owing to its stacking nature and low atomic thickness.[110] Graphene also carries some crucial factor such as mechanical elasticity and flexibility. Furthermore, the diversified graphene properties are procured with different synthetic procedure.[111] On the other hand, the GO is a material which is derived from graphene by incorporating oxygen rich functional group on the graphene layers. The presence of covalently attached oxygenated groups reduces the electrical conductivity which minimize the chance of materials to be used as electrode.[112] But due to electron acceptance ability of oxygenated functional group (present on the GO nanosheet), favors as an active material for TENG.

2.6.1.2. Boron Nitride (BN)

The BN comprises equal amounts of boron and nitrogen. The structural appearances are similar to graphene with alternate boron and nitrogen arrangement which displays a honey comb like structure.[113] The 2D h-BN used as dielectric layer for graphene transistor which improve the device performance impressively[114] due to its high chemical stability and insulation nature. The electrical insulation of h-BN can be tuned by surface modification with appropriate doping, defect and strain engineering.[115] The composite of h-BN with other 2D materials can also improve the TENG performance considerably. Recently, the h-BN/MoS_2 composites shows better electron acceptance ability and peak to peak output voltage in comparison with the individual material.[116]

2.6.1.3. Black Phosphorus (BP)

The BP has certain features such as tunable band gap, plane in anisotropic transport which favor them a suitable candidate for semiconductor application. Contrarily, for triboelectric material it requires better charge storage capability but simultaneously BP predominantly capable to accumulate electron as well. Therefore, the performance of BP material in triboelectric NG can be utilized as an advantage. However, it is not stable at ambient condition that can overcome by hybridizing it with other material such as hydrophobic cellulose, tempo oxidized cellulose and MoS_2.[117]

2.6.1.4. MXenes

The MXenes are 2D materials consist of carbonitrides, nitrides, and transition metal carbides. Recently, the MXenes have been utilized as an electrode material due to its superior electrical conductivity. The MXenes with electronegative functional groups (-Oxygen and –Fluorine group) favor them as a negative triboelectric material. Primarily, the MXenes are procured via selective etching of the A element which is from MAX phases (includes metallically conductive alloys Ti_2AlC, Ti_3AlC_2, and Ti_4AlC_3).[118] However, there is several reports stated that the modification of MXene with other compatible materials have been utilized for TENG device. For example, the MXene is modified with nitrile butadiene rubber,[119] MXene/TiO_2/cellulose nanofiber,[120] and cellulose-reinforced/MXene composite film,[121] and MXene/silicone nanocomposite[122] for TENG applications.

2.6.1.5. Layered MOF/COF

The MOFs/COFs are the crystalline material with high surface area and high pore size volume. More importantly, their structural integrity can be tuned by using different

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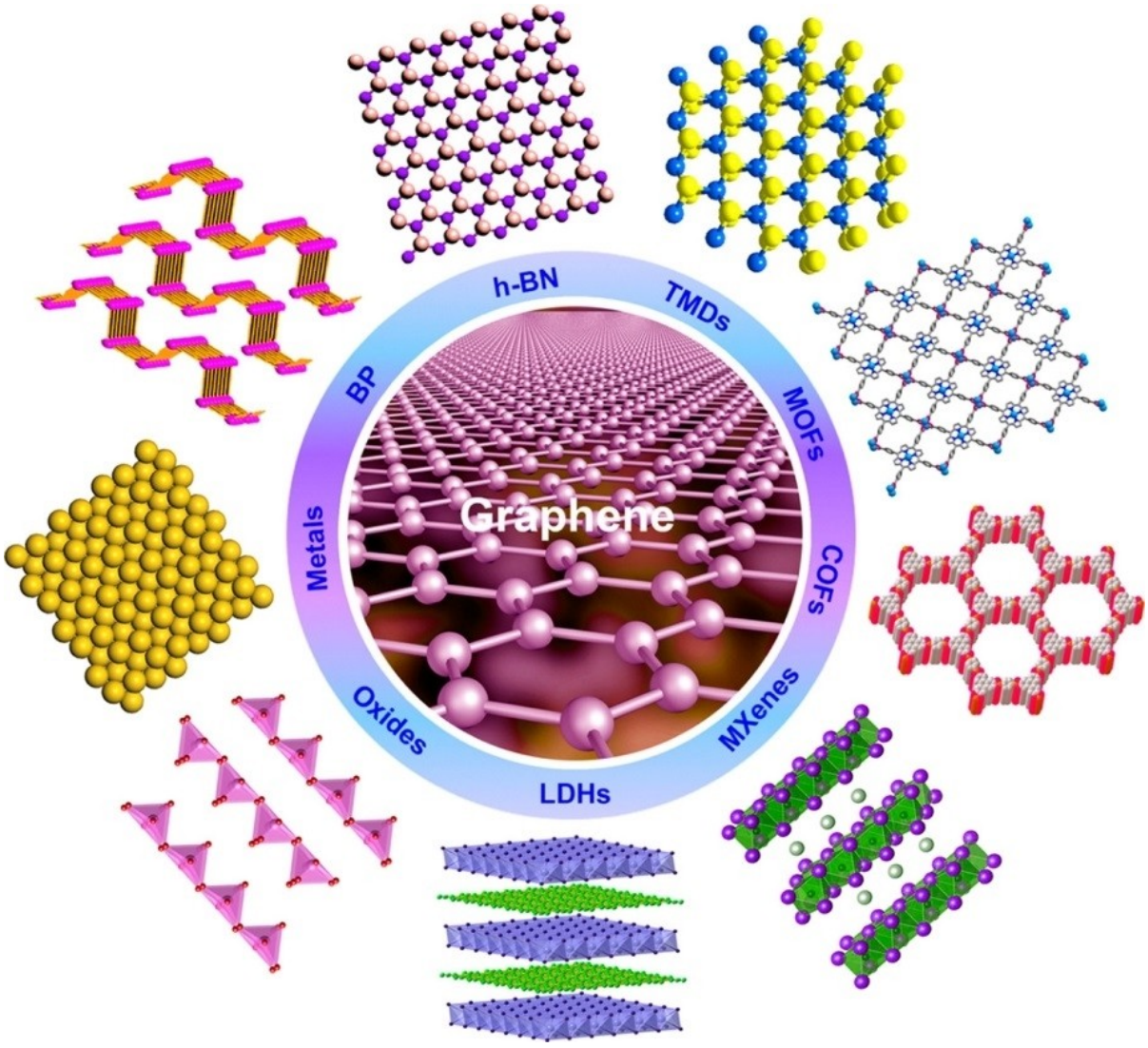


Figure 10. Classification of 2D based materials for TENG.[206] [Adapted with permission from ref. 206 Copyright 2015, American Chemical Society]

linkers.[123,124] Again, the formation of layered structure, it can induce the electron accepting capability which is beneficial for TENG. Furthermore, the layered MOFs display excellent sensitivity to the pollutants and high dielectric constant in humid environments that enhances TENGs capacity to the self-power sensing and stable energy harvesting.[125] Moreover, the cationic COF with electric dipole enhances the TENG performance substantially, because the presence of ionic interface over the COF skeletons.[126] Again, the increment of π-conjugation within the COF skeleton found to be suitable candidate to enhance the TENG performance.[127]

2.6.1.6. Transition Metal Dichalcogenides (TMDs)

The TMDs consist of wide range of materials (WS_2, MoS_2, WSe_2, $MoSe_2$) with wide range of applications such as sensing, catalysis, energy storage, electronics etc.[128–131] Among TMDs, the sulphides are commonly used for TENG application. Despite its excellent properties such as optical, electrical, stretchability, and ultrathin nanostructures, it can also explore to enhance its performance by incorporating ligand conjugation for TENG application. This effect may be due to the electric potential difference between triboelectric layers which was connected with the defect engineering and the p-type doping effect of the TMDs.[132] On the other hand, MoS_2 is found to be capable candidate to minimize the +Ve charge recombination with triboelectric electrons, that can improve

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the output performance.[133] Some composite materials are also found to be highly efficient in TENG application such as polyvinyl chloride/molybdenum disulfide (PVC/MoS_2) membranes. The presence of MoS_2 in the composite films impressively enhance the surface charge density and as a result its wear resistance and output performance improved.[134]

2.6.1.7. Layered Metals

The criteria of materials used for TENG application must be mechanically robust and stretchable in presence of external factor. In this scenario, the layered metals play a crucial role due to its mechanical stability than traditional metals which utilized for suitable wearable devices for human.[135] Consequently, the Au nanosheets are also utilized as an electrode material for TENG, owing to their flexibility and outstanding conductivity.[135]

2.6.2. Different 3D Materials

2.6.2.1. 3D Nanoribbons

The conversion of nanoribbons into 3D wavy structure initiates as flat, thin and crystals of silicon ribbon on mother wafer. Further, the silicon elements are combined with pre strain elastomer substrate. Upon releasing the strain, the Si are converted to wavy stretchable and well controlled nanoribbons.[136] These nanoribbons are highly stretchable and maintained excellent strain up to 10% in case of stretching and compression. Furthermore, the rGO nanoribbons and PVDF were combined to enhance the surface roughness of the thin film. Such materials along with functional groups and surface roughness can promote the charge storage capacity of the thin film and can be utilized as SE-TENG.[137]

2.6.2.2. 3D Nanomembranes

3D nanomembranes can also be engineered by the buckling process. Similar to 3D nanoribbons, the silicon films are converted to the biaxially pre-strained PDMS slab. Upon release of the strained nanomembranes, the controlled surface topology is produced. These membranes are stretchable with controlled morphology at different possible strained system. Therefore, it can be utilized as biaxially stretchable device.[138]

2.6.2.3. 3D Bridge-Island Structures

To achieve a more complex structure, the combination of ribbons with 2D island like structure is possible. The combination of metal oxide and PDMS offers highly stretchable device with strain up to ~140% along with twisting pitch as 90° in ~1 cm.[139] Recently, ZnP nanosheets with 3D laser induced on the graphene foam assembly and utilized as an electrode material. The modified electrode offers high surface area and as a result it enhanced electrical and ionic conductivities.[140]

2.6.2.4. 3D Printing

3D printing process is found to be a viable path to meet the demands of diverse self-powered devices with customized designs in practical applications such as fine-tuned circuits, signal processing, remote controllers, flexible electronics, and active sensor networks. The 3D printing with fused deposition modeling for TENG, it improves the output efficiency by decreasing the Young modulus.[141] The limitations of traditional 3D printing technique, the possibility of micro/nanostructures and thermosetting materials are lower for the fabrication of TENG. The performance of TENG is primarily depend on the structures and properties of the materials. To address these issues, polymer tubes (nylon and tubular polyethylene) are used as carrier boats for direct printing of polyethylene glycol and polydimethylsiloxane without any templates in order to achieve the highest output performance that can be achieved by adding more electronegative material. Also, to fabricate a sponge-like structure after printing by removing of polyethylene glycol which is attached with the film.[142]

2.6.3. Synthetic Procedures of 2D material

2D materials are generally synthesized via top down and bottom-up approaches. Numerous exfoliation processes are commonly utilized for top-down method. The detailed explanations are discussed below:

2.6.3.1. Top-Down Approach

In this approach, the bulk materials are breaking into single or few layer of the materials. This process often produces high quality 2D materials by exfoliation (textural intermolecular forces) of solids with ultrasonication or mechanical stimuli. The multiple layer of exfoliation reduces the intermolecular attraction between the layers and produces exfoliated sheets, liquid-liquid interfaces, polymers etc.[143]

2.6.3..1.1. Micromechanical Exfoliation

The sheets of single or few layers were achieved via micromechanical (ME) process by retaining its original properties and crystalline structure. It is a simple and cost-effective process to produce 2D materials. Graphene is perfect example of 2D material which is produced ME methods with improved conductivity, mechanical strength and optical properties. Further, the exfoliation of TMDs also possible with this

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technique [Figure 11(a)]. But it is not feasible to exfoliate in large scale due to inability to control its layer size and quantity of monolayer surface contamination for sticky polymers.[144]

2.6.3.1.2. Electrochemical Exfoliation

This method allows to electrochemically re-stacking of material layers and produces new material structure is shown in Figure 11(b). From eq. (2), the Li^+ ions are inserted into MoS_2 layers when exfoliated with organic solvents, the hydrolysis of the lithiated material as follows: (eq. (3)).

$$MoS_2 + xLi^+ + xe^- \rightarrow Li_xMoS_2 \quad (2)$$

$$Li_xMoS_2 + xH_2O \rightarrow MoS_2 + \frac{x}{2}H_2 + xLiOH \quad (3)$$

But to achieve the targeted materials for longer time, the careful exfoliation is required. Interestingly, this process can be utilized for large scale production of single layered materials.[146]

2.6.3.1.3. Liquid Exfoliation

Liquid exfoliation is favorable method for preparation 2D materials by intercalation-based exfoliation and direct exfoliation in solvent. Recent reports are available for successful synthesis of monolayer or few layers 2D nanosheets (BP, BN, graphene, TMDs) by sonication in different organic solvents.[146] Further, to exfoliate a single layer TMDs, the electrochemical lithiation process can be carried out. For that, the bulk materials are collected in an electrochemical cell for controlling and monitoring. Upon ultrasonication, the intercalated material in ethanol or water it produces huge amount of single layers of TMDs.[147] Again, selection of suitable solvent is an important aspect of this process. For instance, to exfoliate graphite, the solvent surface tension must be near to 40 mJ/

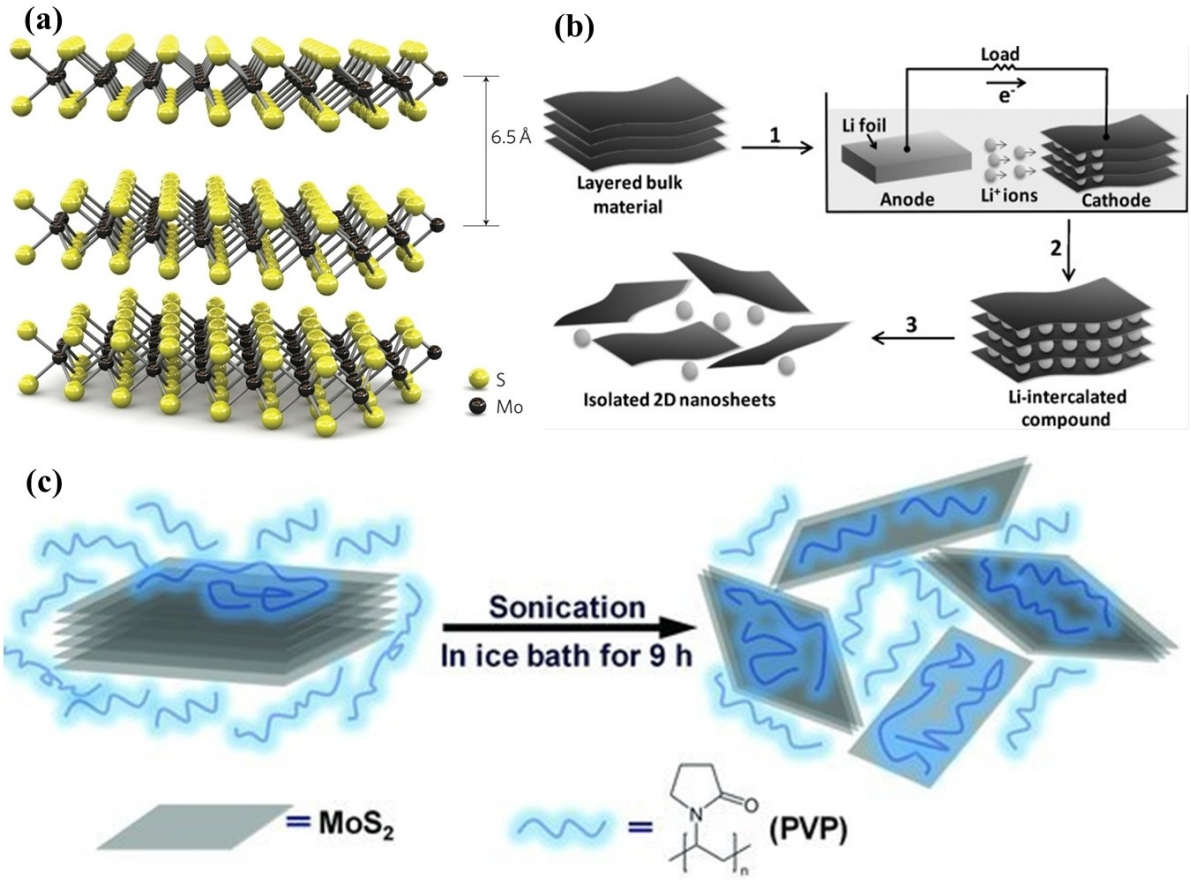


Figure 11. (a) MoS_2 monolayer synthesized by micromechanical exfoliation,[144] [Adapted with permission from ref. 144 Copyright 2011, Springer Nature] (b) Lithium incorporated electrochemical exfoliation of MoS_2,[145] [Adapted with permission from ref. 145 Copyright 2011, John Wiley & Sons] and (c) Illustrative scheme of polymer-assisted exfoliation of MoS_2.[209] [Adapted with permission from ref. 209 Copyright 2012, John Wiley & Sons]

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m^2.[148] However, utilization of mixed solvent strategy, polymer or surfactant may be effective for this limitation.[149–151]

2.6.3.1.4. Ultrasonic Exfoliation

Vander waal forces occurs within the solid when exposed in ultrasonication that converts solid into single or multilayer nanosheets. For this process, the sonication period and choosing suitable solvent is a crucial factor. After exfoliation via sonication, the materials are centrifuged and dried at appropriate temperature. For example, the ultrasonic exfoliation of bulk MoS_2 dried and dispersed in deionized water and, hence the suspension of the materials is obtained [Figure 11(c)]. However, it is suitable for large production through liquid-phase sonication and the resultant sheets can be assembled to the films with its universal properties. But to control the degree of exfoliation is quite difficult.[152]

2.6.3.1.5. Ion-Change Exfoliation

Although, the exfoliation of Vander Waals solids to ultrathin nanostructures can be advantageous, but for ionic solids the strong interaction is present between the layers which is difficult to exfoliate (eg. $LiCoO_2$ and LDHs). The $LiCoO_2$ is a cation exchanged layered metal oxide. Recently, Ju et al. prepared the monolayers of $LiCoO_2$ by treatment of HCl for 3 days and produce cation exchanged $Li_xH_{1-x}CoO_2$, on to which exfoliation with tetrabutylammonium cation intercalation is done. After removal of the unexfoliated particle with the help of centrifugation nanosheets, the powder samples were obtained. Consequently, a nanosheet of thickness ~ 1.2 nm was observed by utilizing electron microscope.[153]

2.6.3.1.6. Lithium-Intercalated Exfoliation

This process mainly produces a single sheet nanostructure of the 2D material. Single layers TMDs are effectively achieved with approximate yield of ~ 100 %. It requires the production of Li_xXS_2 molecule. But some difficulties are associated with this process such as high temperature treatment with longer time is required. Again, to produce the single layered nanosheets, the lithium intercalation should be carried out carefully and with controlled manner. The possibility for the production of metal nanoparticle along with Li_2S should be ruled out. Chhowalla et al. prepared the MoS_2 monolayer by utilizing this technique. The MoS_2 crystals were treated with butyl lithium solution in hexane for the two days of exfoliation. Further, the Li_xMoS_2 is sonicated in water about an hour to avoid deintercalation.[154]

2.6.3.2. Bottom-Up Approach

Single or few layers of the materials are achieved through atoms or molecules from the particular compound. Some materials having stacked bulk crystal which cannot be exfoliate by the above-mentioned process. Bottom-up methods are used to create nanoscale materials from atomic or molecular origins and the effective production of 2D materials is possible.[155]

2.6.3.2.1. Wet Chemical Approach

This approach mainly focuses on the doping of new chemicals during the synthetic steps and it may possible to get smoother surface of the material by using various ligands. The major advantage of this technique is to produce materials at industry scale (e.g., TMDs production). By choosing eco-friendly chemicals and solution may lead to green chemistry synthesis.[156]

2.6.3.2.2. Hydrothermal-Solvothermal Approach

In this method, the reaction temperature is more than the boiling point of the solvent. It is especially useful for precursors with the highest solubility and growth phase stability at high temperature and pressure. This method can be utilized to synthesis both 2D and 3D materials. The solvothermal reaction between sulfur and selenium with ammonium molybdate at 150–180° for 48 hours leads to the formation of single layer of MoS_2 or $MoSe_2$ [Figure 12(a)].[157]

2.6.3.2.3. Microwave-Assisted Approach

The common solvents are utilized for this technique as 1-methyl-2- pyrrolidinone, N, N-dimethylformamide, 1-cyclohexyl-2-pyrrolidinone, and two ionic liquids 1-hexyl-3-methylimidazolium hexafluorophosphate, 1-butyl-3-methylimidazolium hexafluorophosphate. The selection of solvents require two main conditions (i) the effective consumption of the electromagnetic energy, (ii) the dispersion of the materials for a considerable duration/concentration.[158] Organic solvents such as 1-methyl-2-pyrrolidinone, N, N-dimethylformamide, 1-cyclohexyl-2-pyrrolidinone have dielectric constants from 25 to 38.25 and can effectively satisfy the condition (i). Further, the solvents having surface tension ~ 40 mN/m can minimize the enthalpy of mixing and promote exfoliation and reduces agglomeration.

2.6.3.2.4. Topochemical Transformation Approach

Topology is a term which is related to morphology of materials. In topochemical transformation, the resultant product carries similar morphology as the precursor's materials

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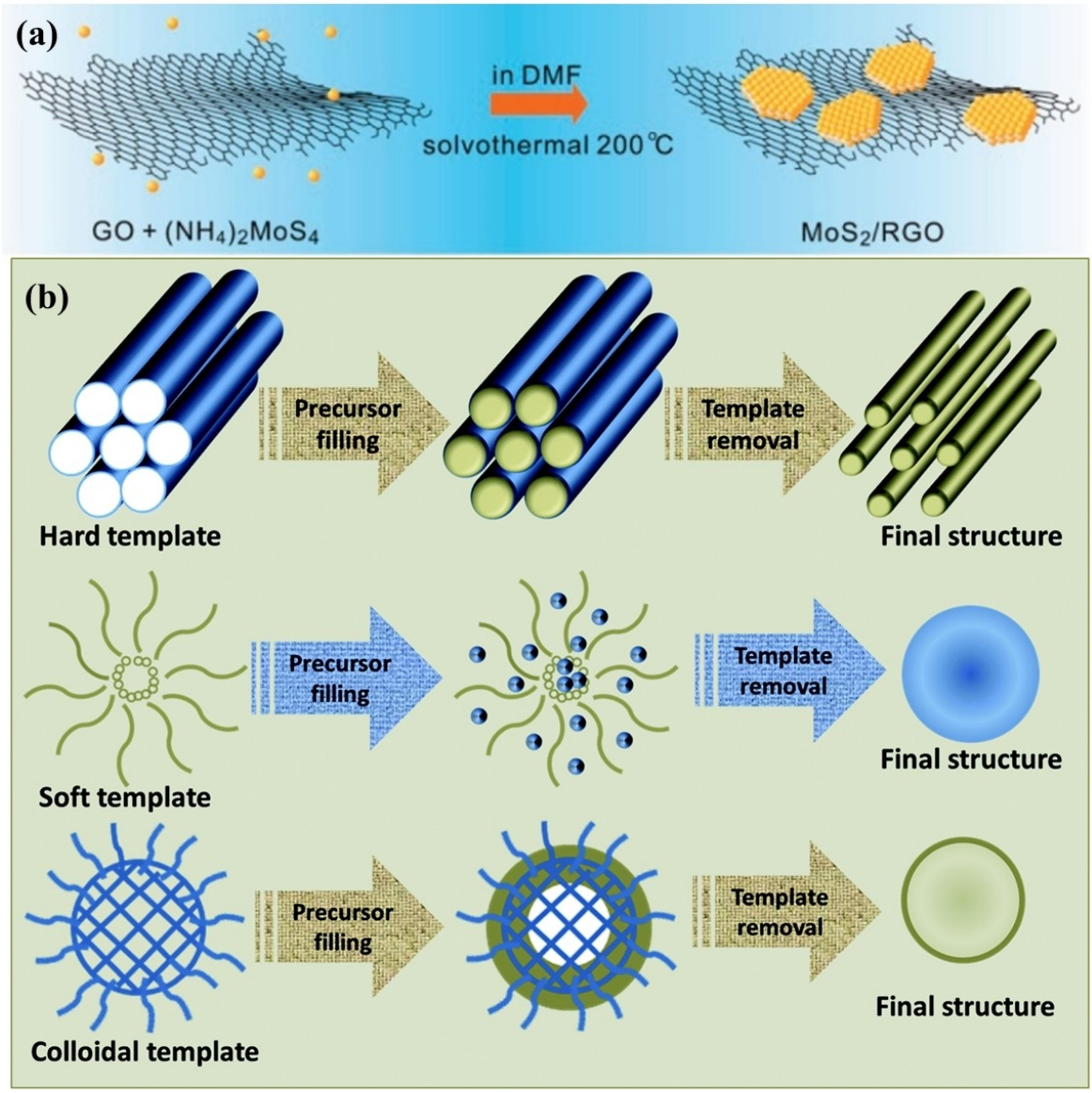


Figure 12. (a) Solvothermal growth of rGO on MoS_2 layer,[207] [Adapted with permission from ref. 207 Copyright 2011, American Chemical Society]and (b) Different templates for synthesis.[208] [Adapted with permission from ref. 208 Copyright 2020, Royal Society of Chemistry]

do. Few materials prepared by this method are δ-FeOOH, CeO_2, and Co_3O_4.[159] Most of the non-layers materials prepared through topotactic oxidation and topo reduction. Recently, the $Ni(OH)_2$ nanosheets were synthesized by utilizing this methods in which ethylene glycol was used as reducing agent and as a result, the similar kind morphology was observed for Ni and $Ni(OH)_2$ nanosheets.[160]

2.6.3.2.5. Template Based Approach

In this method, a template is used during synthesis and it is utilized for nanostructure growth which influences the porosity, and size. Primarily, the template is prepared and utilized for the fabrication of materials. Fabrication can be possible by physical process such as surface coating as well as chemical process (substitution, addition, elimination or may be isomerization reaction). However, the elimination process requires dissolution or harsh condition because the obtained materials are operated at high temperature (calcination).[161] This methods offers controllability over the surface morphol-

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ogy, dimension and structure of the desired product. Template may be different such as hard and soft or colloidal templates [Figure 12(b)]. Most of the common 2D materials TMDs, graphene is act as hard templates. Jiang et al. obtained Pd square nanosheets having (100) facets on the reduced graphene oxide.[162] However, to synthesize single crystalline, the Au membrane use as soft template i.e. polymer-free lamellar hydrogel was used. The resultant membrane was found to be ultrathin, flexible with flat surface due to the slow diffusion of the precursor ions in the confined 2D space.[163]

2.6.3.2.6. Chemical Vapor Deposition (CVD)

This technique allows the preparation of 2D layer materials with uniform thickness, large surface area with insulating surface for the fabrication of various types of devices such as flexible, electronic, and optoelectronic. By maintaining the growth parameter such as temperature, flow rate of carrier gas, precursors, pressure of the reaction chamber, thickness and the domain size of the structure of the material can be controlled. But intermittently complex intermediate reaction may lead to the non-uniform thickness, slow growth rate, and adsorption impurity. To deal with such difficulties space confined growth can be a better choice. By this method, the growth of large graphene single crystals can be obtained with the degradation of precursors CH_4 and by reducing the nucleation density.[164]

2.6.3.2.7. Physical Vapor Deposition (PVD)

In this method, the materials are stored on a surface assisted by heated atomic source. To engineered 2D materials, the PVD process is adjusted with highly purified atomic sources and ultrahigh vacuum conditions. The production of materials such as MoS_2, $MoSe_2$ and monolayer alloys with considerable vapor pressure at high temperature can be vaporized and stored at relatively low temperature. The fabrication of $MoS_{2(1-x)}Se_{2x}$ based monolayers depend on the H_2 flow rate, deposition temperature, temperature gradient, pressure and growth time.[165]

2.6.4. Application of 2D Material

2.6.4.1. 2D Materials as an Electrode for TENG

Nowadays, the demand of flexible and foldable devices is very significant. Such devices carry bending, twisting, folding and stretching capability. Flexible and transparent TENG can be the exciting candidate for such kind of the devices. For highly powered, transparent and flexible TENG requirement, the higher optical transparency and both the electrode (top or bottom) must be active under external stimuli.[166] Chu et al. reported a TENG on human skin that utilizes as wearable device without any requirement charging process.[167] The SE-TENG assembled with polyethylene terephthalate (< 0.9 μm), polydimethylsiloxane (< 1.5 μm) and thin graphene (< 1 nm) working as substrate, electrification layer and electrode, respectively. The TENG on human skin can produce electricity with the help of cloth or various contacts over the body. The effective area of contacts allows to touch the sensors that sends a communication signal from human gesture to digital signals. Recently, Shin et al. fabricated a TENG with the help of graphene (as electrode) and poly(dimethylsiloxane) (as active layer).[168] The electrode material can be transferred to any specific substrates such as glass, fibers polymeric sheet which favors for practical application at low cost. However, some demerits occur by using graphene based TENG. The grain boundaries and wrinkles reduce the conductivity of graphene films and weaker the Vander wall force interaction between substrate and graphene films. To overcome such issues, the graphene is combined with conductive polymers for better output performances. In this aspects, the Yang et al. assembled a CVD grown graphene with poly(3,4-ethylene-dioxy-thiophene):polystyrenesulfonate polymer to prepare a composite which associated with the high transmittance as 83.5% of the sheet resistance of 85 Ω.[169] The composite is capable to enhance the power and current density to 118% and 140%, respectively. Therefore, the composite material displays outstanding performance for TENG by maintaining electrical characteristic even after 10000 cycles.

Apart from graphene, the materials such as MoS_2 and MXene are also explored for TENG application. Lan et al. fabricated TENG by using 2D MoS_2 nanosheets with 1D Ag NWs as composite materials.[133] The sandwich structure resulted, the device showed excellent stretchable (~70%) property with highest gauge factor of 215.4. The combined effect of MoS_2/Ag NWs improve the surface area and reduces the resistance of the films, consequently increases the conductivity and junction area. On the other hand, the MXene carries excellent properties such as high electro-negativity, superior electrical conductivity, therefore combining MXene with Ag NPs, the elastic fiber can significantly able to reduce the fiber resistance.[170] For TENG application, the single fiber acting as electrode and PDMS as triboelectric layer which offers superior flexibility, mechanically robust and stretchability. Therefore, by applying this TENG into cultivation field to protect the crops that could harvest energy from raindrops. Wu et al. fabricated a liquid-SE-based TENG by using GO dispersions (as liquid electrode) and PDMS (as triboelectric).[171] The 2D structure of GO enhances the charge transfer and it achieves a maximum power density of 4.97 W/m^2.

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2.6.4.2. 2D Materials as Triboelectric

The materials such as polymers, silk, metal etc. exhibits triboelectric properties. To confirm the position of 2D materials in triboelectric series, Seol et al. studied the triboelectric charging behavior of various 2D materials.[172] It is found that among 2D materials, the MoS_2 tends to receive more electrons while other 2D materials are lies in the negative region of the series. However, the synthetic procedure and materials thickness does not have that much effect on charging polarity but chemical doping can significantly induce the triboelectric charging properties of MoS_2. Therefore, the doping of MoS_2 with $AuCl_3$ and benzoyl viologen produces p-type and n-type doped MoS_2. Further, the p-type doped MoS_2 enhances the work function and vice-versa in case of n-type doped MoS_2.

The surface area is also a critical factor to get the higher output performance of TENG, numerous approaches have been identified to enhance the triboelectric surface area. Guo et al. utilized GO as triboelectric materials for TENG.[173] The higher surface area, multiple functional group, strong space charge effect of GO facilitate high electric field than the usual negative triboelectric material. Again, the GO can also be an effective component to inhibit the growth of bacteria, therefore, GO based TENG have certain advantages such as electrical performance and antibacterial properties than TENG without GO.[174] In some cases, the TENG performance can be tuned in the molecular level. Zhai et al. synthesized a TPB–DBBA–COF by using 1,3,5-tris (4-aminophenyl) benzene (TPB) and 2,5-dibromobenzene-1,4-dicarbaldehyde (DBBA) as linker.[175] The high surface area along with the electron withdrawing linkers and the presence of bromine groups favor to get the highly efficient COF for TENG application.

For TENG, electrons are the most important aspects for their performance because electrons are responsible for the generation of electricity. Any process leads to decrease the number of electrons will significantly hamper the TENG performance. There are several phenomena to loss triboelectric electron:

i. When two triboelectric surface causes air breakdown due to opposite static charges with high voltage.[176]
ii. The TENG electrons coupled with positively charge ion, or to the particle present in the environment.[177]
iii. The electric field leads to the drift process as well as the diffusion of the electrons, facilitates the loss of electrons.[178]

Xia et al. utilized the spin coating technique to assemble the aligned graphene sheets (AGS) and PDMS [Figure 13(a–b)].[179] Further, the composite AGS@PDMS film have been utilized for TENG. The obtained result of the modified materials is quite interesting, the film enhances the output current up to 3 times than the PDMS as individual. However, the increasing amount of graphene forms the conductive layers, consequently reduces the dielectric properties and overall performance. On the other hand, the MoS_2 is excellent electron withdrawing materials due to its high surface area.

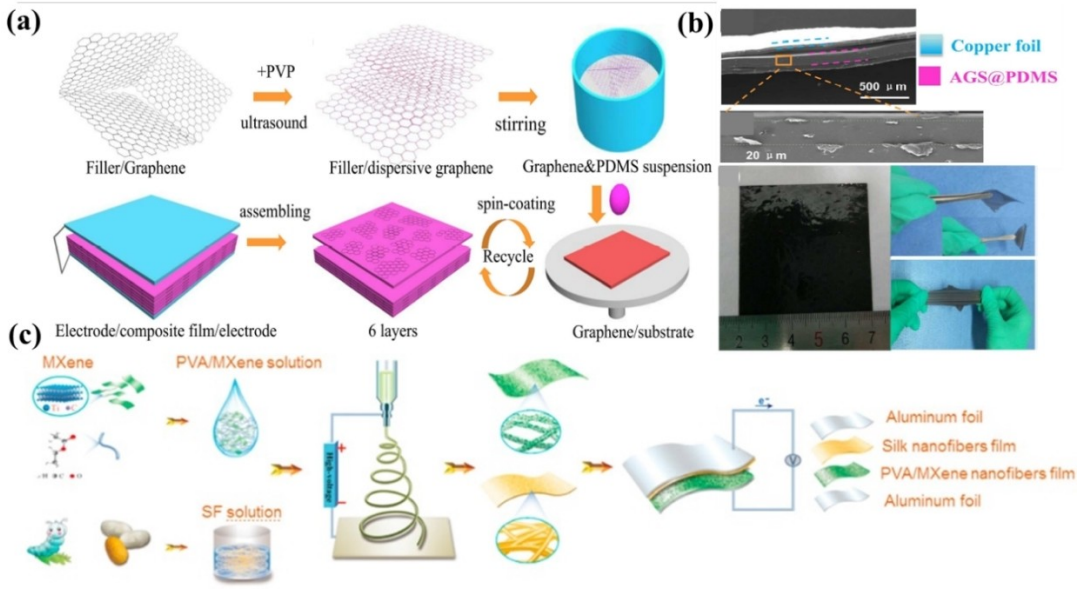


Figure 13. (a, b) Illustrative diagram for fabrication of AGS@PDMS composite film,[179] [Adapted with permission from ref. 179 Copyright 2017, Elsevier]and (c) Fabrication of PVA/MXene nanofiber film, silk nanofiber film, PVA/MXenes, and silk-nanofiber-film-based TENG.[180] [Adapted with permission from ref. 180 Copyright 2019, Elsevier]

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Therefore, Wu et al. assembled the polyimide layer with MoS_2 which can significantly reduce the electron loss possibility.[178] Consequently, the MoS_2 based TENG achieved higher power density as high as 120 times than the without MoS_2. Recently, electrospinning nanofibers are a hot topic for the TENG application. Jiang et al. fabricated poly(vinyl alcohol) (PVA) with MXenes into an electro-spun nanofibers for TENG [Figure 13(c)].[180] This nanofiber improves the TENG performance in comparison to pure PVA film. MXene plays an important role toward the overall performance of the nanofibers in three different ways:

i. The MXenes carries high conductivity, therefore it can significantly reduce the film resistance.
ii. The combined effect of polymer and MXene enhances the surface area, consequently enlarge the effective area of contact.
iii. MXene is having electronegative functional group like – Fluorine and oxygen containing groups which can utilize to prepare the composite film with high electronegativity.

The composite thin films endorse the highest output power density of ~1087.6 mW/m^2 at an input resistance of 5.0 MΩ. However, to synthesize the 2D TENG which functions on harsh environmental conditions such as humidity, temperature, and fire challenging task. Recently, Ahmed et al. synthesized a self-extinguishing triboelectric and fire-retardant NG.[181] The device shows excellent triboelectric properties due to the presence of GO and polyacrylonitrile nanofibers. The fire-retardant NG is inflammable in presence of butane flame even after the exposure of 90 s. This property of fire-retardant NG can lead to a path breaking materials for life saving wearable device.

2.6.4.3. 2D materials as an Electrode and Triboelectric

As discussed earlier, the graphene is having large surface area and excellent conductivity therefore, it has the potential to play a dual role in the TENG application. Kim et al. utilized CVD grown graphene as top electrode as well as triboelectric material for TENG.[182] The surface ripples and wrinkles of graphene enhance the adequate friction, providing excellent output voltage. However, regularly stacked graphene layer promote more output performance in comparison to randomly stacked layer. On the other hand, the Jiang et al. fabricated a highly flexible MXene film with laser-induced graphene (LIG) electrode which act as porous triboelectric material for TENG.[183] Different 2D based nanomaterials are compared with recently published articles in table 2.

The PDMS and MXene assemblies are highly electronegative and porous, which tremendously improve the conductivity. The composite material induces 7th times more output performance than the single PDMS based TENG.

2.6.5. Natural Materials

The natural based material for TENG application is beneficial with respect to sustainability, low cost and safety purpose. Jia et al. reported a Hosta leaf based TENG, where the electrolyte and water present inside the leaf can causes to enhance the conductivity.[184] The leaf based TENG can function for few days and produces current and voltage of 9.5 μA and 230 V at 2 Hz. Along with PMMS, different contact materials were utilized which resulted to charge a capacitor and lit up LEDs. Moder et al. also reported charge accumulation at Rhododendron plant leaf which acts as outer dielectric.[185] The inner tissue of leaf and cuticle layer are accountable for the charge transfer and ionic conductivity. Further, it was investigated the charge generation capability of dry leaf and living leaf and was observed that, the dry leaves were unable to produce the charge and its transfer. The result suggests, the leaf cuticle was responsible for their charge generation process. Further, the Feng et al. fabricated fresh and dry leaf powder as VCS TENG.[186] Both the fresh leaf as well as dry leaf powder with PVDF-PI and poly L-lysine (PLL) produces output voltage of 430 V and 1000 V at 15 μA and 60 μA respectively. However,

Table 2. Comparison of 2D based material for TENG application.

Sr. No.	Material	Modes	Electrode	Triboelectric material	V_{oc} (V)	I_{sc} (μA)	Pressure	Power (mW m^{-2})	Ref.
1.	Graphene	SE	Graphene	PDMS	47.1	7	80 kPa	144	[167]
2.	Graphene	SE	LIG	PDMS	44.5	6.8	8.3 kPa	80.8	[111b]
3.	MoS_2	SE	AgNWs@MoS_2	PDMS	95.8	-	60 kPa	160	[133]
4.	Graphene	VCS	Cu	Cu–graphene@PDMS	117	26	–	4800	[179]
5.	GO	SE	Al	GO	1100	55	125 kPa	3130	[173]
6.	MoS_2	VCS	Al	MoS_2– polypropylene (PP)	7.2 V	0.52	200 kPa	276.3	[116]
7.	BP	VCS	Au	Au–BP@cellulose nanofibrils	5.2 V	18 mA m^{-2}	700 kPa	10.6	[117]
8.	Au NSs	VCS	Au NSs@PDMS	PDMS–Au NSs	98.9 V	2.8	15 kPa	0.3	[135]
9.	MoS_2	VCS	Al	Al–PI@MoS_2@PI	400 V	-	–	25 700	[178]
10.	Graphene	SE	Graphene	PDMS	4.98 V	0.52	–	–	[168]

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Chen et al. VCS TENG with rose petals and PMMA.[187] The output voltage and current for the fresh rose was achieved about 30.6 V and 0.78 μA, which reduced to 23.2 V and 0.5 μA as the petals dried. Moreover, upon removal the wax coating from the rose petals it generates the output voltage and current up to1.83 V and 0.26 μA. Thus, the electrolyte of the petal plays an important role for charge transfer.

An et al. come up with a different approach to fabricate biopolymer-based TENG by utilizing lignin/nylon blend and soy-protein/nylon blend.[188] Copper films with NFs and PI (nanofiber and polyimide) acts as positive and negative electrodes for the ecology TENG. The solution-blown nylon NF (SB-NNF), the TENG deliver higher output values compared to ES-NNF (electrospun-nylon 6 nanofibers) based TENG. The solution-blown soy protein NF (SB-SNF) as well as solution-blown lignin NF (SB-LNF) TENG deliver lesser output than SB-NNF based TENG. Yao et al. fabricated TENG by using cellulose nanofibrils (CNFs) from wood pulp and FEP (fluorinated ethylene propylene) as frictional layers.[189] The fabricated TENG deliver output current and voltage of 7 μA and 5 V, respectively. Further, this TENG are modified with fiberboard into power board which can deliver 10–30 V and 10–90 μA when somebody stepped on it. On the other hand, Kim et al., the fabricated an arch-shaped bio-TENG with the help of bacterial nanocellulose (BNC) (from Acetobacter xylinium KJ1).[190] The device was fabricated by two ways (i) BNC- Bio TENG by using Cu as opposite layer electrode, (ii) Bacterial nano cellulose polyimide (BNC PI)-TENG by using PI/Cu as opposite contact layer. Similarly, the Chandrasekhar et al. fabricated a C-TENG by using cellulose powder and PDMS.[191] It can produce current, power and voltage nearly 2.8 μA, 576 μW and 28 V by utilizing 5% of the composite films. It can be further modified to self-powered locomotion detection and energy harvesting by attaching with floor.

Kim et al. fabricated a TENG with Ag nanowire layer and CNF.[192] This nanowire was further coated on CNF by vacuum filtration method. The fabricated TENG can produce electricity by writing and erasing on an e-paper. The TENG can also be useful in material removal, breath monitoring, and antibacterial activity. He et al. reported a cellulose fiber (CF) based CF-TENG.[193] It has utilized, the cellulose microfiber (CMF) and cellulose nanofiber (CNF) as template to obtain a 2D structure consisting of CMFs/CNFs/Ag paper. Further, the CF-TENG mask was used for breath monitoring and cotton mask for CF- TENG with PM 2.5 removal. The observed breath was found to be 15 and 45 min^{1} for normal and slow running state. Again, it can remove the PM 2.5 with efficiency of 98.83%. Further, the antibacterial study confirms the importance of silver layer and it can efficiently kill the 6 × 107 and 2.4 × 107 CFU/cm^2 for *Staphylococcus aureus* and *Pseudomonas* respectively.

Quin et al. fabricated AP-TENG on Ag/PET substrate where CNF and PDMS inks are used to print the 3D patterned positive and negative layers.[194] Further, it has used freeze drying and post annealing process to enhance the performance of the TENG. However, different contact angle (filament axis direction of CNF and PDMS) displays that, the fabricated TENG can deliver high output performance at 0° in comparison to 45° and 90°. Niu et al. fabricated amino (A) functionalized CNF as A-CNF TENG.[195] The fabricated TENG with FEP as an opposite triboelectric layer delivered output of 13.4 μA, 195 V, and 65 nC. However, upon modification of CNF, the obtained TENG performance is well in humid conditions. Zhang et al. developed regenerated cellulose with dissolution and regeneration process [Figure 14(a)].[196] The overall performance has increased due to the high surface area of the cellulose II aerogel. It was also observed that, by using natural polysaccharides like chitosan and alginic acid significantly improves the performance. This TENG was coupled with socks to harvest energy while walking and running. Wang et al. described a cellulose/PVA hydrogel (CPH) based SE TENG.[197] The voltage and current outputs of the CPH-TENG were found to be 41 V and 500 nA, respectively and it remain stable at 0° to 60° C temperature. Jiang et al. fabricated natural material based BN-TENG by using five different natural materials such as egg white (EW), silk fibroin (SF), chitin, rice paper (RP) and cellulose.[198] With the help of inductively coupled plasma (ICP)-reactive ion etching, the effective area of contact of the biopolymer can be improved. Further, the lifetime of the BN-TENG can be improved with SF encapsulation layer. Moreover, the combination of SF and RP as opposite layers deliver current, power density and output voltage of 0.32 μA, 21.6 mW/m^2 and 34 V respectively. Furthermore, the proposed BN-TENG was shown to regulate the beating of cardiomyocyte clusters C1, C2, and C3. Zhang et al. used RSSP to fabricate TENG [Figure 14(b)].[199] Charge affinity of the protein can be modified by changing genetic sequence of the protein. Further, the water lithography (WL)-based inkjet printing created surface roughness on RSSP. However, RSSP can also be modified by chemical functionalization with Ag, carbon nanotube, graphene, and Au by WL to further improve the performance. Furthermore, the RSSP patch was effective against gram-negative (Escherichia coli) and gram-positive (Staphylococcus aureus) bacteria by creating reactive oxygen species (ROS) within the bacteria. This study proposed a variety of RSS modification procedures, including chemical conjugation, mesoscopic mixing and gene modification to enable its application in adjustable TENG.

Kim and his groups developed lipid-based TENG employing synthetic and lipid layers on the surface of the leaves, hair cells, leaves and skin etc.[200] To understand the role of lipid, the four different synthetic lipids and sample without lipid was

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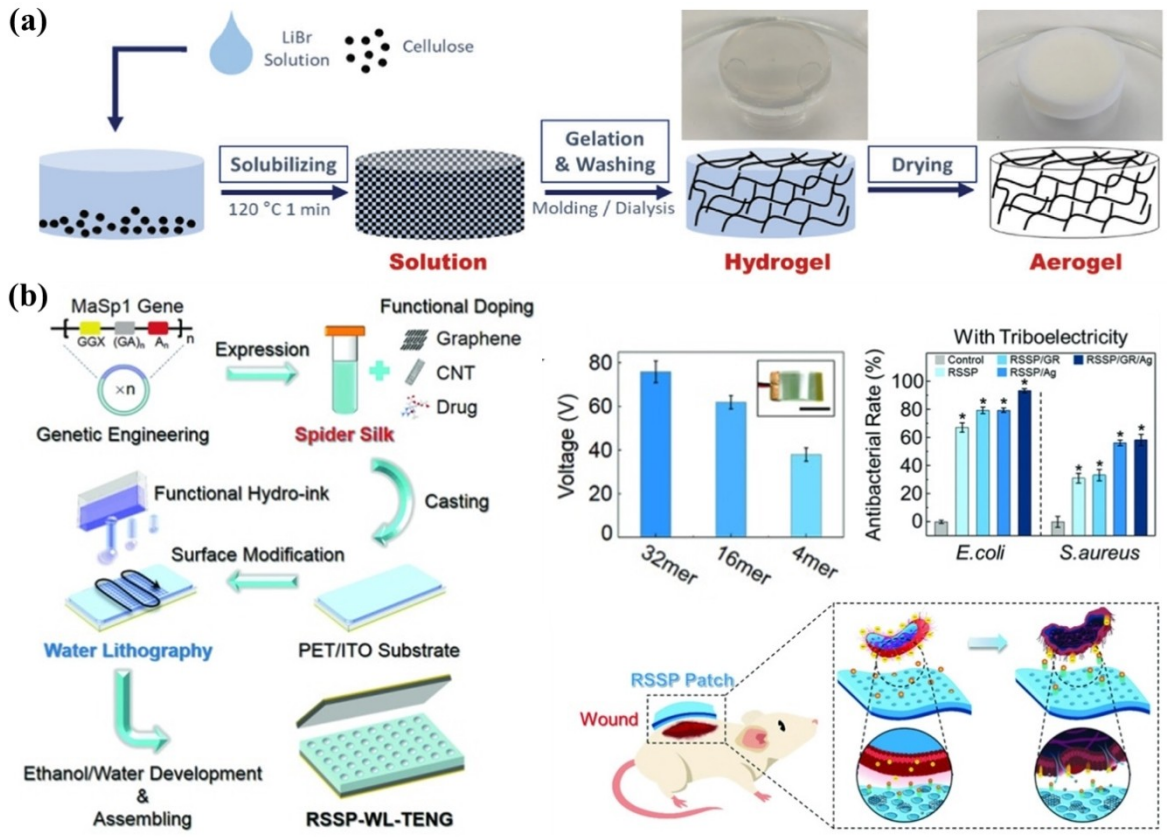


Figure 14. (a) Synthetic scheme of cellulose II aerogel,[196] [Adapted with permission from ref. 196 Copyright 2020, John Wiley & Sons] and (b) Recombinant spider silk protein (RSSP) based TENG, its voltage output and representation as an antibacterial patch.[199] [Adapted with permission from ref. 199 Copyright 2018, John Wiley & Sons]

taken. It was observed that sample without lipid shows comparatively less surface potential than the sample with synthetic and transferred lipid. All the lipids upon analyzed by Kelvin probe force microscopy (KPFM) behaves as positive triboelectric materials. Therefore, the lipids in contact with PTFE delivers ~ 100 times more contact surface potentials. It was also found that the natural and synthetic lipids outperform the Nylon 6 in comparison of surface potential. Additionally, to capture the wind energy, an energy harvesting vine (silicone/conductive fabric/silicone) was wrapped around the branches. Khandelwal and his coworkers reports E-TENG by using edible materials such as silver leaf, rice sheet, laver.[201] This edible product acts as electrode, substrate, and active layer. The fabricated device offers minimum cell viability effect based on MTT assay, DAPI staining and cell imaging. The E-TENG with FEP as the contact material can deliver current and output voltage of 315 nA and 23 V. The practicality of edible materials for scale-up was proved by creating a big device that showed strong silver leaf adhesion even across a vast area. It can also be possible to power up UV, wristwatch, thermo hygrometer and green LEDs by using this E-TENG. Chin and his coworkers use biodegradable rice paper (RP) and conducting ink to fabricate SE mode RP-TENG.[202] The RP-TENG with PVC as contact materials can deliver output current and voltage of 6 μA and 24 V respectively. To further improve the overall performance, it was modified by rice paper powder (RPP) and conductive as RPP TENG. The modified RPP TENG can deliver output current of 16.7 μA and voltage of 392 V.

2.7. Metal-Organic Frameworks (MOFs)

Metal-organic frameworks (MOFs) are a kind of porous material with higher crystallinity, Yaghi and co-workers defined it for the first time in 1995.[210] MOFs are prepared by reticular synthesis process, which involves assembling building blocks or units into a specified ordered network, it consists of metal nodes and organic linkers,[211] as shown in Figure 15. Recently, the MOFs have received a lot of interest as a new type of porous material with a high surface area and persistent porosity. The size and shape of MOFs may be tuned for the TENG applications. Due to their intrinsic features such as

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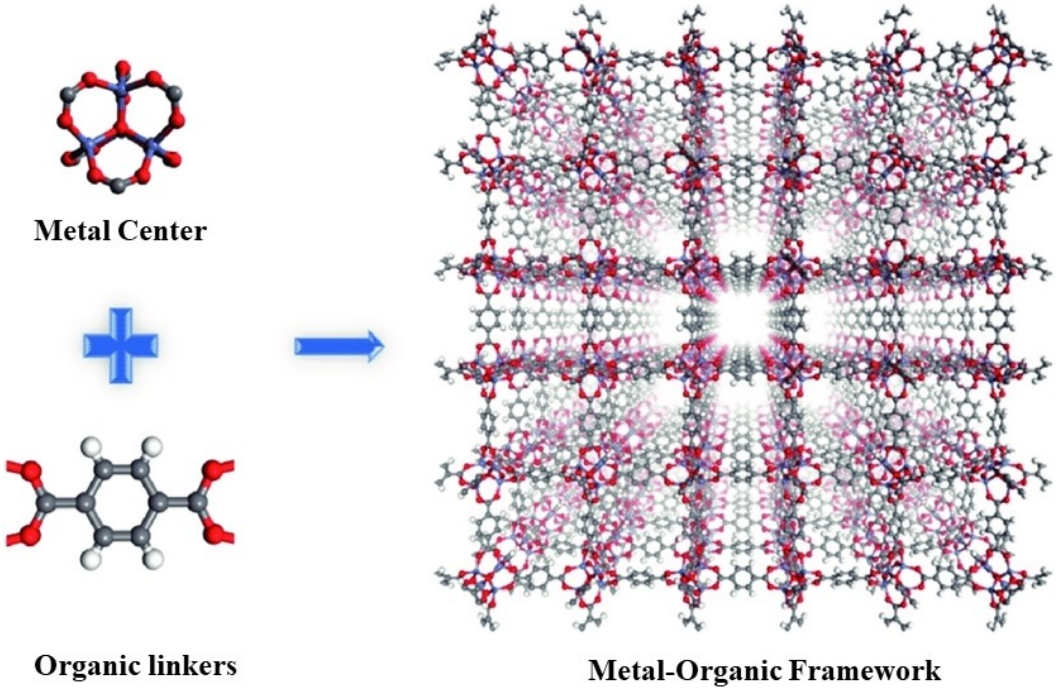


Figure 15. Schematic representation of MOF preparation.[211] [Adapted with permission from ref. 211 Copyright 2022, LCGC North America]

configurable pore diameters, accessible metal sites, controllable crystal structure, morphology, and functionalization of MOFs that led to explored in various applications such as sensing, energy storage, energy conversion, energy harvesting, gas storage, etc.[212,213a,b] The utilization of MOFs in energy harvesting applications such as TENG has emerged as a new field in recent years. MOF in TENGs is more appealing because of their easy surface tunability, which is important during contact and separation. TENG contact area and electrical output are increased by tribo-layers having a wide surface area. In recent years, the researchers have developed MOF-based TENGs to improve electrical performance based on these findings. Khandelwal et al. and their co-workers have been developed TENGs, based on the families of ZIF-based MOFs and utilized as power for low-efficiency electronic devices.[214]

2.7.1. MOF Based Materials

Recently, the MOFs have been investigated for the conversion of mechanical energy into electricity via-TENG which combines the triboelectric effect with electrostatic induction. The MOF offers versatility in terms of TENG size, functionality, and geometry, making them suitable candidates for a variety of applications. The porous nature of the MOF provides large surface area, which helps TENG devices function in better way. The TENGs are used in a variety of applications due to their high output performance, cost-effectiveness, ease of construction, high efficiency, and a wide range of materials.[215]

Khandelwal et.al prepared zeolitic imidazole framework (ZIF)-8 based MOF for TENG as well as a self-powered tetracycline sensor. In the study of ZIF-8, the framework has achieved remarkable chemical and thermal stability, due to consistent porosity for the MOF-based TENG device. The expansion of ZIF-8 was validated by structural and morphological investigations. The KPFM was used to determine the ZIF-8 surface potential and confirmed its positive triboelectric behavior. The MOF-TENG device was fabricated in the typical VCS method, using positive and negative triboelectric layers of ZIF-8 and Kapton, respectively. The output performance of MOF-TENG was investigated using detailed electrical characterization. In VCS mode, the MOF-TENG achieved output voltage of 164 V at 7 μA. Finally, the MOF-TENG is used to successfully design and demonstrate a self-powered UV counterfeit system as well as a tetracycline sensor. The sensor is highly-rated selective and can be washed and reused.[216]

Jayababu et al. firstly prepared bimetallic (Co/Zn-MOF), herewith Zn was chosen to incorporate in cobalt-MOF and achieved efficient output. The bimetallic MOF has elliptical nanosheets-like morphology that leads to an increase in the surface area that supports the contact and separation mode as shown in Figure 16(a–c). The prepared bimetallic MOF were coated on flexible conductive fabrics (FCF) were used as tribo-

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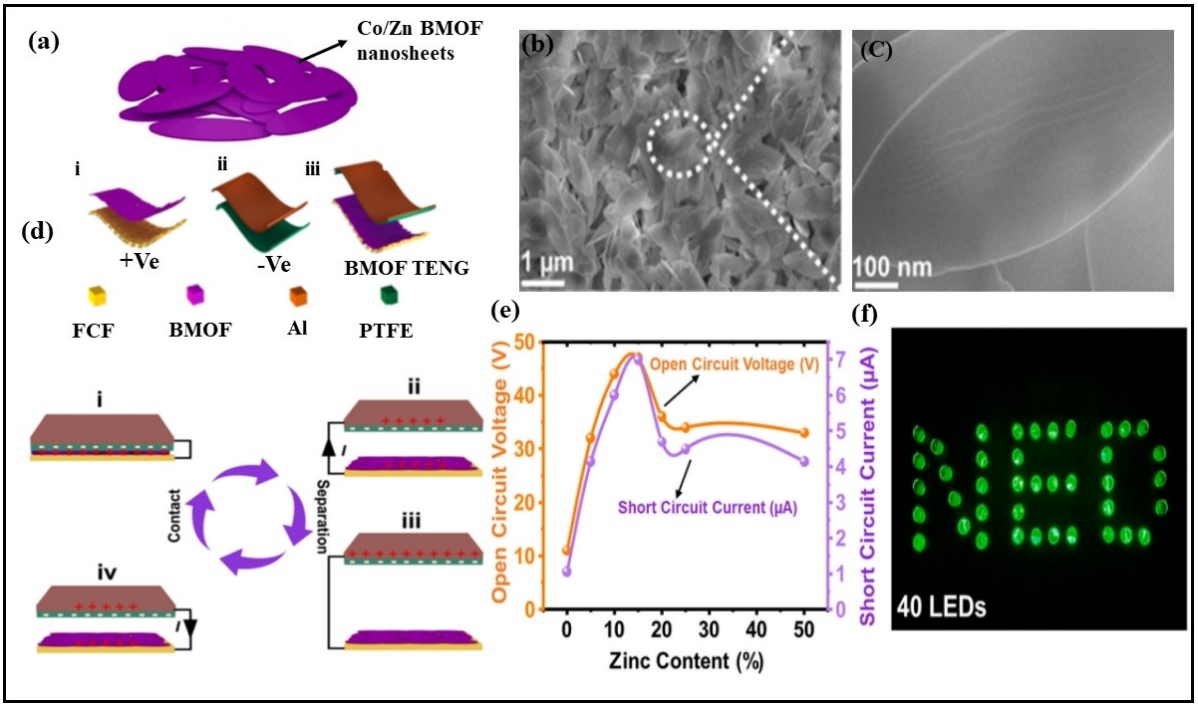


Figure 16. (a) Schematic representation of Co/Zn bimetallic MOF, (b, c) FESEM images having sheet-like morphology, (d) Internal schematic and working mechanism of bimetallic MOF-TENG, (e) Voc & Isc with different Zn loading content, and (f) LEDs illumination with bimetallic MOF-TENG device.[217] . [Adapted with permission from ref. 217 Copyright 2021, Elsevier]

positive material and polytetrafluoroethylene (PTFE)/Al foil was used as tribo-negative material by varying the concentrations of zinc (Zn) from 0 to 50% for achieving the maximum output as shown in Figure 16(d–e). The performance obtained at 0–15% content of Zn, which is higher with Voc is 11 V to 47 V, Isc is 1.06 μA to 7 μA with a charge density of 4 nC/cm^2 to 17 nC/cm^2. The output power of the TENG device is achieved 1.1 Mw/cm^2 with a load resistance of 2 M'Ω that is employed for gas sensing (ammonia sensor) at room temperature without the use of an external source of electricity.[217] Guo, et al. prepared Fluorinated (F)–MOF as a dual-functional to create highly efficient TENGs. The F-MOF leads to an increase in the charge separation due to doping of the fluorine atoms that helps in creating the surface roughness which is a crucial factor for efficient TENG devices. Further, it was prepared on Al foil and polydimethylsiloxane@F-MOF composites as a triboelectric pair, the prepared materials achieved a maximum output Voc of 530 V and Isc of 3.2 μA and reaches a higher power density (52 μW/cm^2) which is 11 times higher than normal TENG. Further, the TENG device could illuminate 40 commercial LEDs as shown in Figure 16(f). The efficiency of TENG is also increased by changing different kinds of polymers (such as polyacrylonitrile, polyurethane, Si-rubber). The fabricated F-TENGs were used to enlighten liquid-crystal displays (LCDs).[218]

Hajra et.al prepared ZIF-67 at room temperature by solvent-assisted route having rhombic dodecahedral morphology. Moreover, the simple contact mode (S-TENG) was fabricated whereas ZIF-67 was used as a tribo-positive layer while polydimethylsiloxane (PDMS/Teflon) as a tribo-negative layer and achieved a maximum output of 118 V with a current of 1.7 μA with power density of 15 μW/cm^2. The S-TENG device was connected with a robotic gripper and performance was measured by changing different objects. Various gaits (jumping, walking, jogging, and running) were recorded by assembling the S-TENG device on both sides of the flip-flop. The triboelectric series will be expanded with ZIF-67 as a suitable contender for designing TENG with a variety of self-powered applications. The ZIF-67 maintains its original morphology after being attached with electrodes.[219]

Shaukat et al. prepared MOF-5, which is having high chemical, thermal stability having rod-like morphology and tunable porosity shows outstanding properties in the fabricated TENG. Whereas, the MOF-5 acts as tribo- positive layer and polytetrafluoroethylene (PTFE) acts as a tribo-negative layer in TENG and achieved a maximum output Voc is 484 V while Isc achieved 40 μA with an efficient power of 3,174 μW. The TENG provides maximum outputs under harsh environmental conditions such as humidity and temperature. The fabricated TENG device further powered commercial LEDs, capacitors,

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and low voltage electronic devices such as a stopwatch and calculator.[220]

The UIO-66 NH_2 based MOF was explored for TENG device by Wang et al. due to controllable size, high porosity, and having octahedral surface morphology. For charge transfer and energy generation in a TENG, the contact area and electric potential of the surface are critical. Herein, UIO-66 NH_2@polydimethylsiloxane composites were used as contact separation mode in which composite exhibits 4 times higher output voltage and 60 times higher current as compared to polydimethylsiloxane based TENG.[221]

Khandelwal et al. synthesized ZIF-62 by using mixed linker strategy and was confirmed by morphological analysis having cubic-like shape which was further explored for TENG devices. The ZIF-62 act as positive potential while Teflon, ethyl cellulose (EC), and Kapton act as negative material. The as fabricated TENG device achieved a maximum output voltage of 62 V while the current is 1.4 μA and the charge is 14 nC. The ZIF-62/TENG device follows the order of TENG/Teflon > Kapton > EC, acting as positive material and generating low output voltage in combination with ZIF-62.[222]

Khandelwal et al. synthesized MIL-88 A for TENG devices by solvothermal route having high porosity and size tunability without modifications in topology which is having rod-like shape. Whereas, MIL-88 A act as tribo-positive and FEP, Teflon and Kapton act as tribo-negative material. The MIL-88 A has an output voltage of 88 V with a current of 2.2 μA. It follows the order TENG, FPE > Kapton > EC. The fabricated TENG device is used to detect hand, leg motions, and low-powered electronic devices. Fluorinated ethylene propylene is a highly negative material and achieves maximum output.[223] Li et al. has prepared a surface-attached metal-organic framework (SURMOF) by cycling spraying methods. The prepared SURMOF coated on the different substrates including plant leaves silk, cotton, etc. The covalent linkage formed between the SURMOF and substrate. The SUMOF layer on plant leaves use to determine the plant physiological activities and environmental behavior. Further, the TENG devices are used to determine the self-driven sensing of wind speed.[224] Li et al. prepared methyl orange doped polypyrole@-ZIF-8 cellulose fiber (ZIF-8/MO-PPy@CelF) to fabricate the TENG. Herewith, the ZIF-8/MO-PPy@CelF acts as tribo-positive behavior and PTFE acts as tribo-negative behavior that could create VCS mode. Further, the cellulose morphology plays a crucial role to enhance the contact separation that leads to an increase in the TENG performance and achieved a maximum Voc output of 129 V and Isc of 6.8 μA. The prepared TENG is used to illuminate LEDs, self-powered supercapacitors etc.[225] Chen et al. prepared a series of metal-ion coupled MOF having ZUT-75(Cu), ZUT-75(Co), ZUT-75(Zn), ZUT-75(Mn) with a VCS mode by using poly-(vinylidene fluoride) PVDF, as tribo-negative material whereas ZUT-75 based MOF as tribo-positive behavior, whereas Kapton as a storage layer. The PVDF has better toughness and resistance that leads to longer durability during friction in TENG. All metal ion species show better efficiency and charge density which followed the order as ZUT-75(Co) > ZUT-75(Zn) > 75(Cu) > ZUT-75(Mn). Among them, cobalt exhibits a higher charge density of 127.05 $\mu C/m^2$, with a power density of 3280.50 mW/m^2, respectively.[226] Wen et.al prepared HKUST-1@PDMS act as tribo- negative and Cu acted as positive respectively. By altering the percentage ratio of HKUST-1 from 0–5 % there will be a change in the performance of TENG devices. The HKUST-1 with a 5 % ratio achieved maximum power of 3.17 mW with a load resistance of 10 MΩ with a relative humidity of 10 % and 13 % which is more efficient than bare PDMS. The more efficient humidity resistance performance is achieved due to the absorption of MOF with water molecules. Herein, MOF provides a novel strategy in TENG devices with efficient output in a highly humid environment.[227]

Hajra et al. synthesized cyclodextrin-based MOF(CD-MOF) which is biocompatible and efficient material. The CD-MOF (alpha (α), beta (β), gamma (γ) is applied in energy harvesting TENG devices. (α, β, γ) CD acted as an organic ligand for CD-based MOF preparation. The surface potential plays a crucial role in output performance. Herein, the CD-MOF acted as a tribo-positive while Teflon acted as tribo-negative, it follows the order A/CD-MOF > B/CD-MOF > G/CD-MOF and achieved a maximum device output of 152 V, 1.2 μA. The fabricated TENG is utilized to power the devices such as capacitors, calculators, etc.[228]

2.8. Coordination Polymer (CPs)

The CPs are composed of porous crystalline material which constituted metal ions and an organic ligand. The wide range of CPs or MOFs materials has led to their widespread use in the field of energy storage, conversion, and harvesting.[229a] The terms CPs and MOFs are used mutually to achieve effective compiste material. The term CP was coined in 1950 whereas, the MOF was coined in 1995 by Omar yagi and co-workers. The CP encompasses the structure when metal ions and organic ligands are joined together to form a sheet, 3D architecture, or an infinite chain. For 1D or 2D networks, the MOF term is often regarded as inappropriate. The CPs based materials show outstanding performance by tuning the concentration of metal ion and organic ligands. To overcome the difficulty in modification of traditional materials like functionalization, tuning in structure or morphology. As an electrode material, CPs or MOF-based materials have been used in fabricating TENG devices.[229b,230] Huang et al. prepared a series of cadmium (Cd-CPs) by controlling the electronic structure and geometry such as (binuclear, trinu-

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clear, pentanuclear clusters) which are named CdCP-1, CdCP-2, CdCP-3.The performance of the TENG devices was analyzed and achieved a maximum output. It follows the order CdCP-1 > CdCP-2 > CdCP-3 based on the nuclear cluster whereas, the binuclear cluster lost electrons very rapidly. The Voc performance of CdCP-1 is 548 V, with a Isc is 78.2 μA, whereas, CdCP-2 having Voc of 502 V, Isc of 69.5 μA, and CdCP-3 having Voc of 453 V and Isc 55 μA at 5 Hz with a dimension of 5 cm TENG devices. It demonstrate the performance of the device decreases with an increase in nuclear cluster numbers.[231]

Wang et al. prepared Cu(I)–CP with different coordinated sites (Cl, I, Br) to determine the performance of TENG devices by altering the different electronegativity terminal groups. The size, shape, polarity, and 3D morphology of the TENG device play a crucial role in the output performance. Further, the fabricated CPs-TENG devices, act as tribo-positive material, and PVDF act as tribo-negative material. It works on the principle of triboelectrification and electrostatic induction. The performance of the fabricated CPs-TENG (Cl, Br, I) devices at 5 Hz frequency with Voc and Isc of 305 V, 41.3 μA and 375 V, 44.5 μA, and 445 V, 69.4 μA for Cl, Br, I/CP-TENG, respectively. The output of the I/CP-TENG device is the highest, as expected, and is compatible with the molecular polarity, as evidenced by the dielectric constants. To summarize, the output performance of the Cu(I) based TENGs follows the order which is opposite of the halogen electronegativity (Cl, Br, I). To carry out the durability and stability of the I/CP-TENG device the cyclic stability was performed and noticed there is no change after 50,000 cycles. Further, there is no change in the morphology of CPs even after the cycling test which shows the Cu-CPs are highly stable.[232]

2.9. Covalent Organic Frameworks (COFs)

The COFs, are another class of porous crystalline material with lower density than MOFs. The COFs having organic building components can be incorporated into regular networks atomically and reliably. In 2005 Yaghi and coworkers defined it for the first time.[233a] COFs are made of light materials having (H, C, N, B, O, Si) allowing them to be lightweight while maintaining good stability and porosity which is connected via a covalent linkage that leads to better stability. The COFs are extremely simple to change and can be customize. Due to their higher stability nature and porosity, it is used in a broad range of applications like gas adsorption, catalysis energy conversion, energy storage, and energy harvesting.[233b] So far, COFs have been used in frequently to construct devices that directly transform mechanical energy into electrical energy. Due to the 2D sheet-like morphology of COFs and layered network, their charge transport capability is faster in energy harvesting TENG devices that leading to an increase in the output performance of the device.[229,234]

Zhai et al. prepared TFP-DB-COF via imine condensation by using 1,3,5-tri formyl phloroglucinol (TFP) and dimidium-bromide (DB) as a neutral and cationic linker respectively. Another neutral TFP-DP-COF was also synthesized by using 3,8-diamino-6-phenylphenanthridine (DP) as a linker via solvothermal route as shown in Figure 17(a). The prepared COF applied for energy harvesting devices by using PVDF as a tribo-negative layer and TFP-DP-COF and TFP-DB-COF tribo-positive. The performance of these two was analyzed and achieved Isc of 23.5 μA with Voc of 365 V at 5 Hz for TFP-DP-COF and Isc of 47.9 μA with Voc of 815 V at 5 Hz for TFP-DB-COF as shown in Figure 17(b–c). For a cationic TFP-DB-COF based TENG device, the Isc and Voc values were enhanced from 5.2 to 82.2 μA and 315 to 1000 V by altering the working frequency. Figure 17(d), the fabricated TENG used to enlighten a green (1388 LEDs) and red-colored LEDs. In addition, an energy storage capacitor and the constructed COFs-TENG (TFP-DB-CO) device can be used to charge low-power electronic devices such as stop watch and self-powered components.[235]

Zhai et.al prepared bromine functionalized COFs via solvothermal route. The prepared COF has sheet-like morphology in the micrometer range that leads to an increase in the output performance. The prepared COF is used as a tribo-positive layer and PVDF as a tribo-negative layer for creating TENG devices. The bromine functionalized COF has porosity and crystallinity due to the electron-withdrawing group attached to it that leads to resonance effect as compared to bare COF. Further, the fabricated TENG device estimates an Voc of 318 V with Isc of 21.6 μA at 5 Hz frequency for bare COF and Voc of 416 V, and Isc of 43.6 μA for functionalized COF. The long-term cycling performance was performed for 500,000 cycles to check the durability of the device and it also can operate up to a wide frequency range which could be suitable for the practical utility to harvest energy from walking human motion, riding, etc. In addition, the fabricated TENG device could also illuminate 1358 green-colored commercial LEDs. This advancement led to an increase in future COF-based TENG devices.[236]

Various MOFs/CPs/COFs based TENG device are compared with recent literature in table 3.

2.10. Perovskite Materials

The photovoltaic performance of organic and inorganic halide perovskite materials has achieved the maximum power conversion efficiency of 25%. Due to their facile synthesis techniques, adjustable band gap, extended carrier lifetimes, lead free perovskite materials and diffusion length, the perovskite has become an appealing materials.[237–242] Numer-

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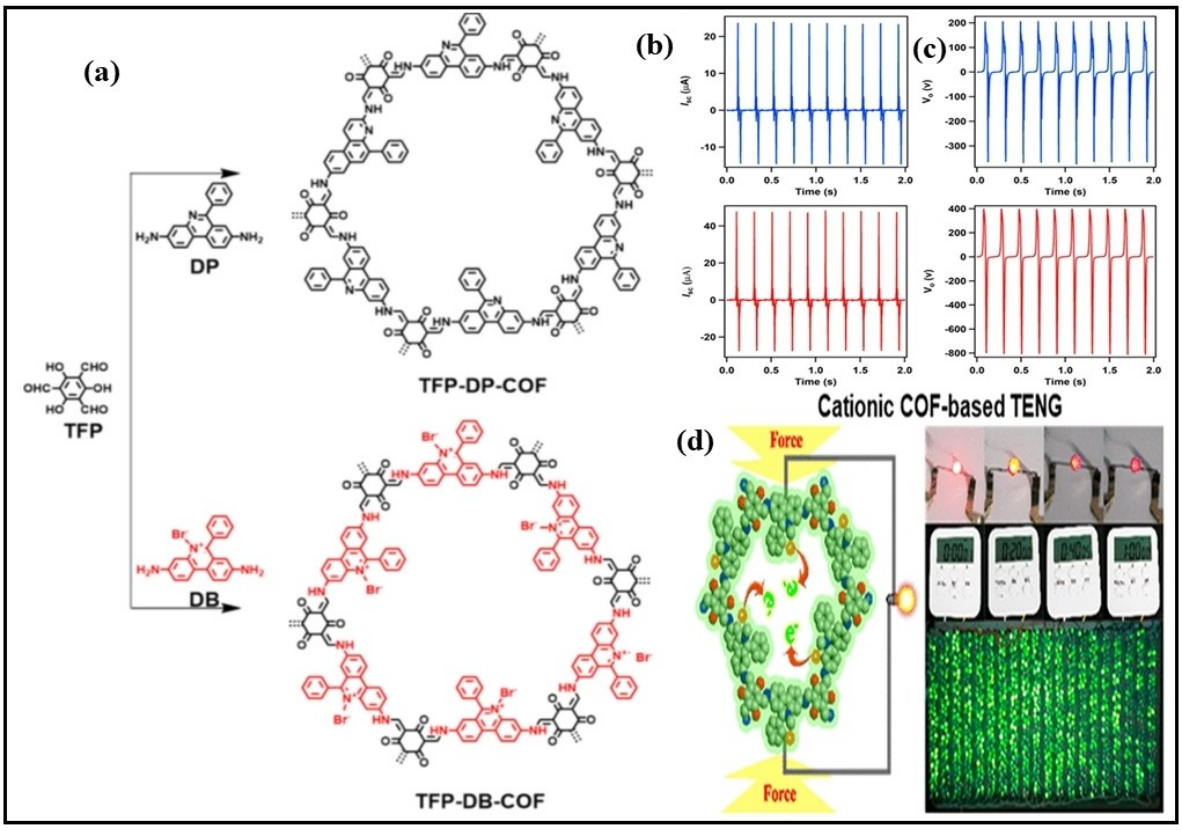


Figure 17. (a) Schematic illustration for the synthesis of COFs, (b) Isc and Voc of TFP-DP-COF, (c) Isc and Voc of TFP-DB-COF, and (d) Photograph of 1388 green-colored LEDs illumination.[235] [Adapted with permission from ref. 235 Copyright 2020, American Chemical Society]

Table 3. MOFs/CPs/COFs based TENG devices and their performance.

Sr. No.	TENG Device	Tribo-Positive material	Tribo-Negative material	Voltage/Power output/current	Power Density/Power/charge density	Ref.
1.	ZIF-62 TENG	ZIF-62	Teflon	62 V&1.4 μA	9.68 mW/m^2	[222]
2.	MIL-88 A TENG	MIL-88 A	FEP	80 V &2.2 μA	16.2 mW/m^2	[223]
3.	MOF-5 TENG	MOF-5	PTFE	27 V & 2.9 μA	5.76 μW	[220]
4.	ZIF-8 TENG	ZIF-8	Kapton	164 V & 7 μA	392 mW/m^2	[216]
5.	Co/Zn-MOF TENG	Co/Zn-MOF	PTFE	47 V & 7 μA	17 nC/cm^2	[217]
6.	F-MOF TENG	F-MOF	PDMS	530 V & 3.2 μA	52 μW/cm^2	[218]
7.	UIO-66-NH_2 TENG	UIO-66-NH_2	PDMS	1.52 mW	1.69 W/m^2	[221]
8.	ZIF-8/MO-PPy@CelF-TENG	ZIF-8/MO-PPy@CelF	PTFE	129 V & 6.8 μA	47.4 nC	[225]
9.	ZUT75-MOF TENG	ZUT-75(Co)	PVDF	565.67 V & 90.72 μA	127.05μC/m^2	[226]
10.	CD-MOF TENG	CD-MOF	Teflon	152 V &1.2 μA	14.3 nC	[228]
11.	HKUST-1 TENG	Cu	PDMS	205 V &37 μA	3.17 mW	[227]
12.	Cd-CPs TENG	Cd-CP-1	PVDF	548 V& 78.2 μA	132.8 μC/m^2	[231]
13.	Cu(I)–CP TENG	I/CP-TENG	PVDF	445 V& 69.4 μA	97.5 μC/m^2	[232]
14.	COF-TENG	TFP-DP-COF	PVDF	365 V &23.5 μA	36.1 μC/m^2	[235]
15.	COF-TENG	TFP-DB-COF	PVDF	815 V & 47.9 μA	73.5 μC/m^2	[235]
16.	B-COF TENG	B-COF	PVDF	416 V & 43.6 μA	51.2 μC/m^2	[235]

ous devices, such as NG, LEDs, transistors, and photodetectors are fabricated by utilizing perovskite materials.[243–246] Perovskites consist of general formula of ABX_3, in which A and B represents the cation ion and metal ion, respectively, and X

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represents halide ion. The scientific communities are exploring to develop renewable energy which can be best suited replacement for fossil fuels. Among energy storage systems, the mechanical harvesting can be a good alternative for fossil fuels and will be a long-term energy source.[247–249] These halide perovskites have recently been used in the creation of NG. Halide perovskites are suitable for NG manufacturing because of their inspirational qualities towards piezoelectric and dielectric nature. Because of the processes of electrostatic induction and triboelectric effect, the fabrication of TENGs exhibit significant benefits in the conversion of mechanical to electrical energy. The easy manufacturing process, high efficiency, various materials and construction possibilities, Wang's group developed the TENG model in 2012.[250,251] Wang et al. used noble metal nanoparticles to adjust the characteristics of a perovskite material ($CsPbBr_3$).[252] To boost the charge storage, the scientists created a bulk Schottky junction, which resulted in a significant shift in NG output. When compared to the controlled device (without Schottky junction), there was a roughly 61% improvement. The solution process deposition is used to fabricate the perovskite material on Fluorine dopped tin oxide (FTO) glass. To prepare the metal/perovskite layer, several metal ion solutions were spin coated onto the perovskite film. The Au, Pt, and Ag were obtained from their metal sources $HAuCl_4$, H_2PtCl_6, and $AgCH_3COO$, respectively. Afterwards, TENG were assembled from the metal film/perovskite and PVDF/Al film assisted as the positive electrode and counterpart electrode respectively. Silver tapes were used to link FTO and Al for device performance through electric wire. TENGs output was capable of lighting roughly 30 LEDs, with a power output of around 3.31 W/m^2 (Table 4).

Tang et al. synthesized an all-inorganic perovskite material with cesium as the cation component. The influence of halide ions (Cl^-, I^-) in $CsPbBr_3$ on triboelectric performance was examined.[253] Using a multi-step deposition spin coating process, perovskite materials were coated on FTO glass and composition of various halide ions was changed. Tang et al. fabricated TENGs perovskite in a VCS mode and tested it for several doping compositions. As positive and negative triboelectric materials, perovskite/FTO glass and PVDF were used. Because, the FTO served as both a support and an electrode for perovskite and silver tape act as electrode for PVDF. However, the TENGs had a total contact area of 7.5 cm^2 (Table 4). When compared to other halide ions, the Br^-, I^-, are the champion composition of perovskite $CsPbCl_3$ which produced the maximum power output. An excellent stability was also observed throughout the course of 60 days. The usage of different halide ions in perovskite produced excellent results in TENGs, which encourages the researchers to pursue perovskite-based TENGs for mechanical energy harvesting.

The impact of barium ion (Ba^{2+}) doping in $CsPbBr_3$ perovskite on surface potential, dielectric characteristics, and electron binding energy of the TENG was optimized by Yudi et al.[254] TENG performance was improved with such perovskite with the lattice structure $CsPb_{1-X}Ba_XBr_3$ (x = 0.01–0.13). The FTO glass was used to deposit a perovskite layer by using a multi-step solution processing approach. The crystalline nature and purity of perovskite materials were revealed by the XRD peak pattern in Figure 18**(b)**. The SEM analysis was performed to examine the exact morphology gained by perovskite materials in Figure 18**(c)**, with an inset crystal lattice, was shown. The TENGs entire motion displacement and output results were depicted in Figure 18**(d–j)**, respectively. A power density of 3.07 W/m^2 was achieved, which was sufficient to illuminate approximately 80 LED lights (Table 4). The TENGs were fabricated by sandwiching perovskite materials-PVDF between the FTO and silver (Ag) electrodes in a VCS mode, with a total active area of 8.1 cm^2 (Figure 18). The effect of humidity and temperature on the TENG was also investigated, but no significant differences in performance were found, demonstrating the TENG's stability and longevity. Based on the overall effect and influence of doping, temperature, and humidity, the fabricated TENG with Ba^{2+} dopped $CsPbBr_3$ perovskite shown exceptional adaptability and reversibility. This work raises the prospect of metal ion doping influencing the performance of perovskite-based TENGs for mechanical energy harvesting. Tang and co-workers[255] explored the doping of several alkaline earth metal ions (M=Mg^{2+}, Ca^{2+}, Sr^{2+}, Ba^{2+}, X = 0–1) as $CsPb_{1-X}M_XBr_3$ in perovskite materials as part of their ongoing study on the doping metal ion in perovskite materials. The deposition of

Table 4. Comparison of Perovskite based TENGs.

Sr. No.	Positive electrode	Negative electrode	Surface area (cm^2)	Power output (W/m^2)	Voc (V)	Jsc ($\mu A/cm^2$)	Ref.
1.	Metal/$CsPbBr_3$–Au	PVDF/Al	6	3.31	240	4.13	[252]
2.	$CsPbCl_3$	PVDF	7.5	3.04	257	87.87 μA	[253]
3.	$CsPb_{0.91}Ba_{0.09}Br_3$	PVDF	8.1	3.07	220	28.8	[254]
4.	$CsPbBr_{2.6}I_{0.4}$	PVDF	9	1.2	192	16.7	[256]
5.	$MAPbI_xCl_{3-x}$	PTFE/Al	0.7	7.99	22.8	0.92	[257]
6.	$Co(OH)(CO_3)_{0.5}$/Pt/$CsPbIBr_2$ layer	PVDF	-	2.04	243	3.1	[258]

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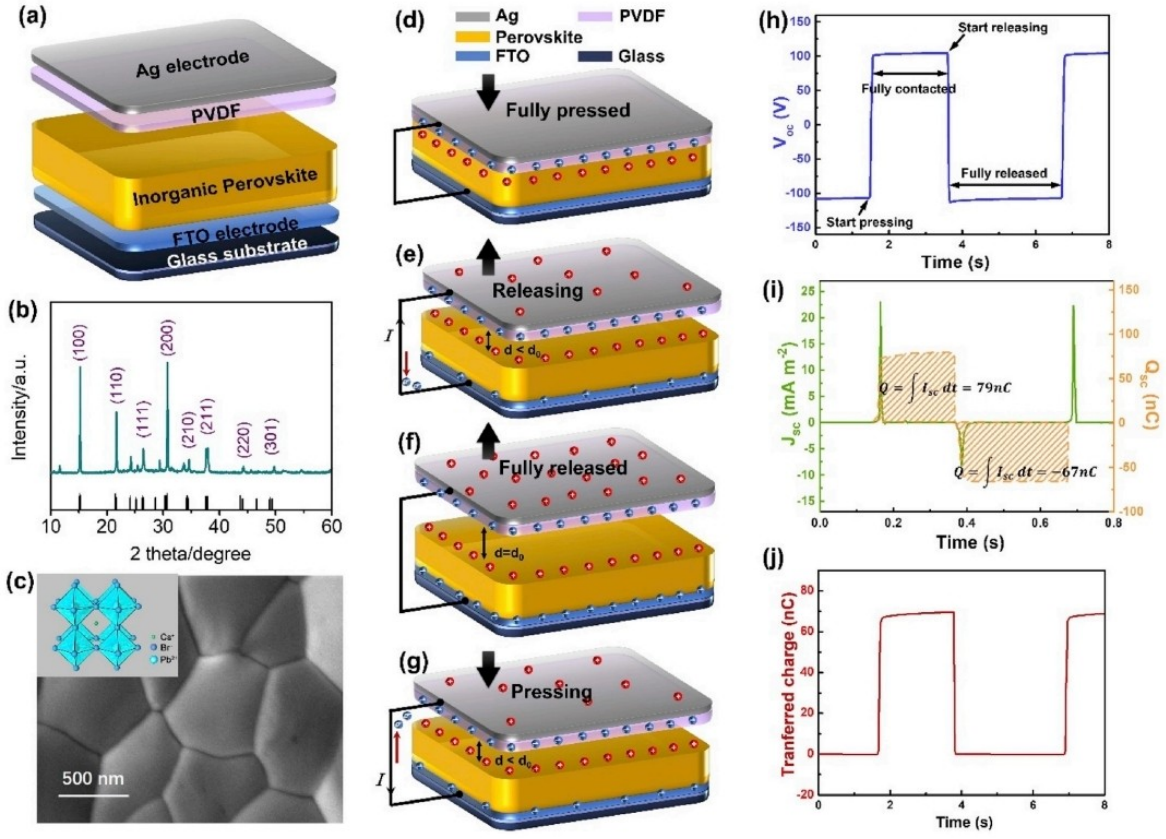


Figure 18. (a) Schematic diagram of TENG (dielectric-on-perovskite) with VCS mode, (b) XRD patterns of the standard $CsPbBr_3$ crystal and prepared $CsPbBr_3$ film, (c) Morphology (SEM image) of $CsPbBr_3$ perovskite film and crystal lattice (inset). Complete "press-release" cycle of the TENG described by motion displacement d: (d) fully contacted, (e) start releasing, (f) fully released, and (g) start pressing, (h) Voc (i) Jsc, and Qsc, and (j) transferred charges output generation during cycle.[254] [Adapted with permission from ref. 254 Copyright 2020, Elsevier]

perovskite materials with or without doping, as well as the assembly of TENG, were carried out in the same way as in prior work. The observed comparison of output performance such as Voc, Isc, and Qsc of different metal ions doped TENGs have been depicted in Figure 19. The electron binding energy, work function, dielectric property, and bond energy of doping alkaline earth metal in perovskite materials were examined to determine the precise sequence of performance. With three distinct configurations under light and dark, the influence of photovoltaic and coupling of electrification was

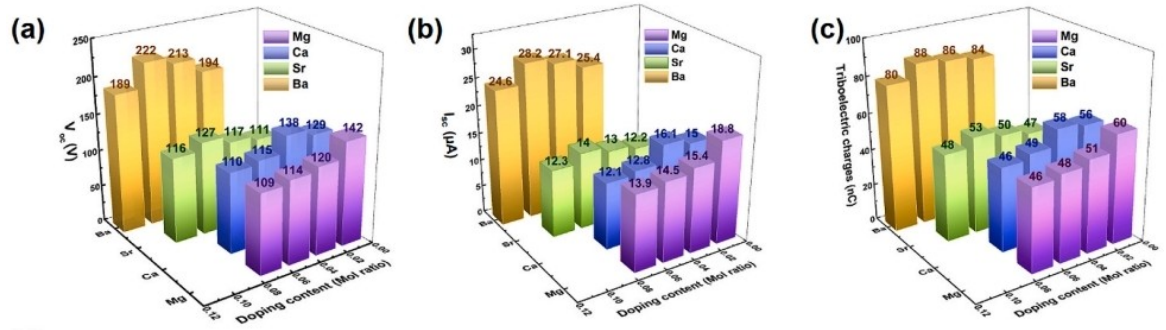


Figure 19. Generated (a) Voc, (b) Isc, and (c) Qsc observations of TENGs based on perovskites ($CsPb_{1-x}M_xBr_3$) with alkaline earth metal doping contents.[255] [Adapted with permission from ref. 255 Copyright 2020, Elsevier]

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also explored. With photoinduced charged TENG, the results demonstrated a significant increase in triboelectrification charges. When two effects are coupled, a reversed voltage signal appears, as well as numerous current signals, indicating the influence of triboelectrification and photoinduced charge.

This study may give a new perspective on the triboelectric series, as well as the foundation for the theory of tribo-photovoltaics and application of photodetectors. Du et al. manipulated the composition of halide ions (Cl^-, I^-) in $CsPbBr_3$ perovskite materials to modify their morphology and dielectric performance.[256] The perovskite was prepared by using a solvent etching approach Figure 20(a). The Figure 20(b–f) represented a rise in perovskite grain size as the DMF/methanol ratio varied, as the higher concentration of DMF, the particle size diameter increases (the relationship between DMF and particle size is provided in Figure 20(g).

The solvent perovskite was employed as a triboelectric material in the manufacturing of TENGs since it was synthesized. The TENGs were fabricated with FTO/perovskite and PVDF films adhered to Ag tape for the positive and negative electrodes, respectively. The contact surface is 9 cm^2 and the distance between the positive and negative electrodes is 5 mm. The TENGs performance was evaluated using a VCS method. Surface engineering has resulted in an increase in charge density and dielectric characteristics. The highest efficiency was obtained in the $CsPbBr_{2.6}I_{0.4}$ composition, with 1.2 W/m^2 (Table 4). Furthermore, the TENG based on perovskite has been demonstrated to be a viable power source to glow the LED lights. This research revealed that, the compositions of halide ions in perovskite may alter their dielectric characteristics. This study paves the way for gathering mechanical energy and converting it to electricity. Song et al. revised a method for capturing solar energy and vibrating energy at the same time.[257] The (Methylammonium = MA) $MAPbI_xCl_{3-x}$ perovskite-based TENG with ETL and hole transport layer (HTL) benefits from a combination of photoelectric and triboelectric effects [Figure 21(a)]. It was the first time the TiO_2 and pentacene were used as ETL and HTL, respectively. Because pentacene has strong air stability, it was grown onto a perovskite layer using a thermal evaporation process. Figure 21(b) shows the TENG's whole functioning mechanism, which is based on perovskite

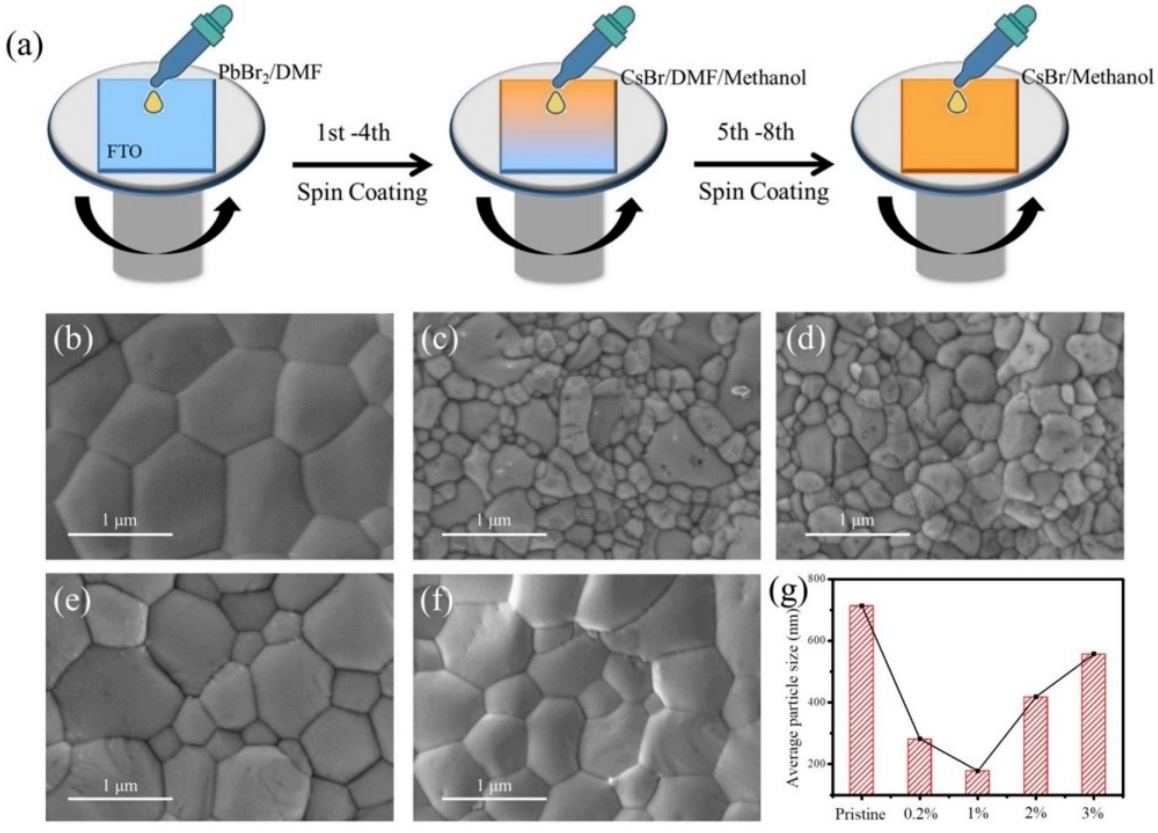


Figure 20. (a) Schematic description of perovskite ($CsPbBr_3$) films through spin-coating deposition technique with solvent etching. Surface review SEM images of (b) pristine $CsPbBr_3$ and etched by (c) 0.2% (d) 1%, (e) 2% and (f) 3% DMF, and (g) Grain size distribution of perovskite.[256] [Adapted with permission from ref. 256 Copyright 2020, Elsevier]

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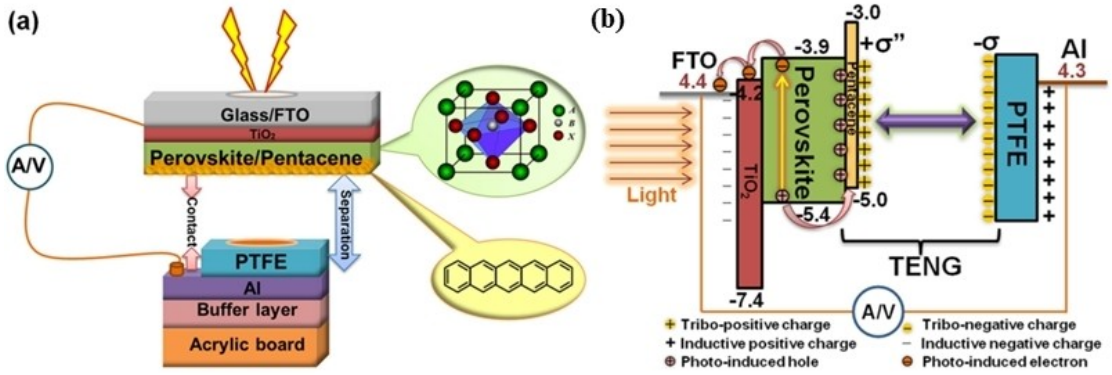


Figure 21. (a) The representation of the optimal P-TENG (b) The working method of P-TENG under illumination.[257] [Adapted with permission from ref. 257 Copyright 2019, Springer Nature]

which includes ETL and HTL. When light falls on perovskite material, it generates a significant number of electron and hole pairs, which travel towards the ETL while positively charged holes stay on the perovskite layer. The TENGs use perovskite materials as positive electrodes, with induced positive charge holes provided as a supplement, resulting in a higher surface charge density. As a result, P-TENGs performance under light was boosted (Table 4). The experiments revealed an increase in TENG performance of approximately 50 % in Voc, and Isc, and charge transfer when compared to a device without HTL and ETL, which had just 0.7 cm^2 active area.

Jian et al.[258] used a multilayer nanoarrays as a friction layer connected to a PVDF layer, which is also beneficial for preventing obstruction when combined with the generated positive charge of electrodes. The interaction of Co-$(OH)(CO_3)_{0.5}$/Pt/$CsPbIBr_2$ and PVDF efficiently increased charge, and the constructed TENG produced better results that quickly illuminated practically all commercial 61 LEDs. Rhombus $Co(OH)(CO_3)_{0.5}$ nanoarrays were deposited on the FTO using a hydrothermal process in which the FTO was submerged in the reaction mixture for 2.5 h at 120 °C. Using a three-electrode system with H_2PtCl_6, the Pt precursor and FTO as the working electrode, a Pt layer was formed on the nanoarrays coated with FTO. After that, FTO was thoroughly cleaned with isopropanol to remove any Pt residue.

Perovskite materials were dropped on the FTO/Co-$(OH)(CO_3)_{0.5}$/Pt using one-step techniques. $Co(OH)(CO_3)_{0.5}$ was evenly dispersed, had a rhombus-shaped morphology and vertically grown on FTO glass, as observed by SEM and cross-section SEM micrographs Figure 22(a–b). In comparison to pristine $CsPbIBr_2$, Figure 22(c–d) shows the deposition of perovskite on nanoarrays (rhombus) with larger effective contact area and surface roughness. For TENG manufacturing, the FTO/$Co(OH)(CO_3)_{0.5}$/Pt/$CsPbIBr_2$ and PVDF on Ag tape were constructed as positive and negative triboelectric electrodes in vertical-contact separation. Finally, the output performance was examined, and it was discovered that the power density was 4.5 times greater with the combination of nanoarrays than with pure $CsPbIBr_2$, which was 2.04 W/m^2 (Table 4). This perovskite-based TENG also demonstrated outstanding heat and moisture stability.

3. Summary and Future Perspectives

So far, we have discussed the detailed investigation of 0D, 1D, 2D, 3D, MOFs, CPs, COFs, and Perovskite materials, starting

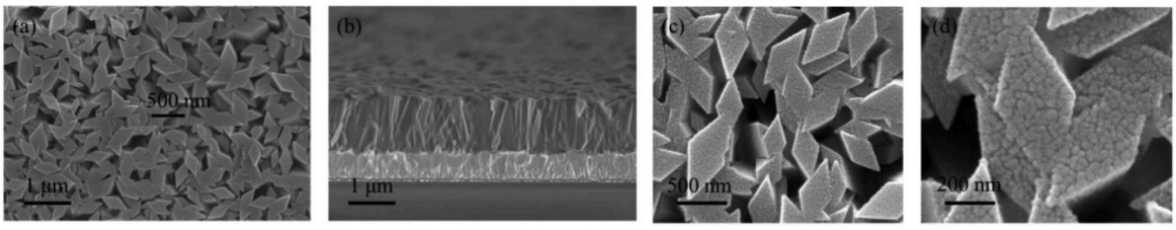


Figure 22. (a) Surface-review and (b) cross-sectional SEM images [$Co(OH)(CO_3)_{0.5}$] precursor. (c–d) surface-review SEM images of 0.25 M $CsPbIBr_2$/Pt-1 with 500 and 200 nm magnification.[258] [Adapted with permission from ref. 258 Copyright 2020, Elsevier]

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from their classification and their application in TENG. However, these materials are found to be perfect TENG materials in terms of durability, flexibility along with energy harvesting, self-powered sensors, and biological based applications. Although these materials achieved tremendous progress in the recent years, there are some challenges to further development and application of this TENG. The recent development of MOFs, COFs, CPs and perovskite based TENGs for outstanding outputs performance due to high surface area and persistent porosity that can be explored for energy harvesting applications such as TENG that has emerged as a new field in recent years. These porous materials in TENGs are more appealing because of their easy surface tunability, which is important during contact and separations in TENGs fabrication. With the controlled morphology synthesis, cost-effectiveness, ease of construction, high efficiency, these findings show that the constructed (MOFs, CPs, COFs and perovskite) TENG devices can be used to charge low-power electronic devices as a self-power resource. Future prospective can be underlined as follows:

i. Ultrathin different dimensional based nanomaterials are explored well in the recent years but many of them are need to be explored yet.[203] To improve the performance these materials can be synthesized with desired thickness, size as well as crystal phase etc.
ii. Some of the TENG's are sensitive to external factor like temperature, humidity etc. Therefore, these material can be modified with other stable material to survive such extreme conditions.[181]
iii. Low impedance device is an important aspect of TENG, generally TENG devices have high impedance. Therefore, to design the low impedance materials with high stability and output should be the major concern in the near future.[204]
iv. The understanding of detailed investigation of the mechanism behind high output performance is important. For example, surface modification of triboelectric layers with small organic molecules improves the charge density of the materials.[205]
v. Knowledge of simulation study is also important to develop stable, high output, low impedance and eco-friendly materials for TENG.

4. Abbreviations

TENG	Triboelectric Nanogenerators
NTSC	National Television System Committee
VCS	Vertical Contact-Separation
SE	Single Electrode
AC	Alternating Current
FSTL	Free-Standing Triboelectric Layer
LBD	Langmuir-Blodgett Deposition
LED	Light-Emitting Diode
FTO	Fluorine Doped Tin Oxide
Voc	Open Circuit Voltage
ETL	Electron Transport Layer
HTL	Hole Transport Layer
PVDF	Poly- (Vinylidene Fluoride)
AC	Alternating Current
QDs	Quantum Dots
CQDs	Carbon Quantum Dots
CNT	Carbon Nanotubes
GO	Graphene Oxide
BP	Black Phosphorus
MOFs	Metal-Organic Frameworks
ZIF	Zeolitic Imidazole Framework
FCF	Flexible Conductive Fabrics
PTFE	Polytetrafluoroethylene
PDMS	Polytetrafluoroethylene
LCDs	Liquid-Crystal Displays
EC	Ethyl cellulose
SURMOF	Surface-Attached Metal-Organic Framework
I_{sc}	Isc-Short Circuit Current
CD	Cyclodextrin
CPs	Coordination Polymers
COFs	Covalent Organic Frameworks
TFP	1,3,5-Tri Formyl Phloroglucinol
DB	Dimidiumbromide
DP	3,8-Diamino-6-Phenylphenanthridine
BN	Boron Nitride
BGE	Bandgap Energy
AuNPs	Gold Nanoparticles
LSPR	Localized Surface Plasmon Resonance
FETs	Field-Effect Transistors
NWs	Nanowires
SWCNTs	Single Wall Carbon Nanotube
PDMS	Polydimethylsiloxane
RMS	Reverse Micelle Solution
PANI	Polyaniline
MEMS	Microelectromechanical System
CVD	Chemical Vapor Deposition
PET	Polythene Terephthalate
ZnO NWs	Zno Nanowires
VLS	Vapor-Liquid-Solid
AgNWs	Silver Nanowires
RPP	Rice Paper Powder
MTT	3-[4,5-dimethylthiazol-2-yl]-2,5 diphenyl tetrazolium bromide
ICP	Inductive Coupled Plasma
RSSP	Recombinant Spider Silk Protein
DMF	Di-Methyl Formamide
CPH	Cellulose/Polyvinyl Alchol Hydrogel

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IE	Ion Etching
WL	Water Lithography
ROS	Reactive Oxygen Species
KPFM	Kelvin Probe Force Microscopy
RP	Rice Paper
EDL	Electrical Double Layer
QLED	Quantum Light Emitting Diode
PVC	Poly-(Vinyl Chloride)
AGS	Aligned Graphene Sheet
OLEDS	Organic Light-Emitting Diodes
AFM	Atomic Force Microscope
VLS	Vapor-Liquid-Solid
TMDs	Transition Metal Dichalcogenides
LPE	Liquid Phase Exfoliation
LIG	Laser-Induced Graphene
ES-NNF	Electrospun-Nylon 6 Nanofibers
SB-SNF	Solution-Blown Soy Protein NF
CNFs	Cellulose Nanofibrils
CMFs	Cellulose Microfiber
PI	Polyimide Nanofibers
NF	Nanofiber
FEP	Fluorinated-Ethylene Propylene
PCF	Poly(Caprolactone)Fibres
PMMA	Poly (Methyl Methacrylate)
USE	U-Shape Electrode
SEM	Scanning Electron Microscopy
PXRD	Powder X-Ray Diffraction
IOTs	Internet Of Things
AC	Alternating Current
0D	Zero-Dimensional
1D	One-Dimensional
2D	Two-Dimensional
3D	Three-Dimensional
PVA	Poly(Vinyl Alcohol
CF	Cellulose Fiber
FA	Fabricated Amino
EW	Egg White
SF	Silk Fibroin
RP	Rice Paper
MA	Methylammonium
BNC	Bacterial Nanocellulose
PLL	Poly L-Lysine
SB-LNF	Solution-Blown Lignin Nano Fiber
DBBA	2,5-Dibromobenzene-1,4-Dicarbaldehyde
TPB	1,3,5-Tris (4-Aminophenyl) Benzene
PVD	Physical Vapor Deposition
DAPI	4', 6-diamidino-2-phenylindole
NG	Nanogenerator

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